Multiobjective Optimization-Based Hyperspectral Band Selection for Target Detection

Meiping Song, *Member, IEEE*, Shihui Liu, Dayong Xu, and Haoyang Yu, *Member, IEEE*

***Abstract*— Band selection can reduce information redundancy and improve application efficiency of hyperspectral images (HSIs), such as classification. Traditional band selection methods usually weight different objectives and combine them together in a single function, making it difficult to balance conflict between various criteria. Recently, multiobjective optimization band selection (MOBS) techniques have got a lot of attention, which evaluate bands from different aspects separately and search for a solution well-balancing all the objectives simultaneously. However, few algorithms are focused on target detection, and most of them use just two objectives, which are not sufficient enough to select optimal band subset for detection. To alleviate these problems, this article proposes an algorithm named target-oriented multiobjective optimization of band selection (TOMOBS). First, the multiobjective optimization (MO) framework with three objective functions is modeled, involving information, noise, and correlation of the bands, respectively. Second, to optimize the proposed model, the noninferior solution advantage matrix (NAM) is designed based on the swarm intelligence optimization method, which can provide accurate solutions and improve the descriptiveness of MO problems (MOPs). Third, a target-oriented evaluation mechanism is developed to guide selecting final result from the Pareto front (PF), especially designed for target detection. Experiments on real hyperspectral datasets show that this algorithm can provide a subset of bands with strong representational capability for target detection and achieve impressing results compared with the state-of-the-art methods.**

***Index Terms*— Band selection, hyperspectral images (HSIs), multiobjective optimization (MO), target detection.**

I. Introduction

HYPERSPECTRAL images (HSIs) provide a huge amount of spectral information about the scene, which is important for remote sensing image processing [1]–[4]. However, abundant spectral information represented by hundreds of bands contains much redundant. Therefore, it is necessary to reduce the dimension of HSIs' bands. There are two types of HSI dimensionality reduction techniques: feature extraction [5]–[7] and band selection [8], [9]. Feature extraction keeps the main information of the inherent data in the new feature space through effective mapping, but the physical information of the inherent data is lost during the mapping process. Band selection is a dimensionality reduction strategy that chooses fewer decisive bands while maintaining efficacy. Therefore, band selection is a promising method for HSI processing, which not just helps reduce the dimension of data but improves the efficiency of HSI analysis.

Band selection is usually implemented in a supervised or unsupervised manner. The supervised band selection is based on prior knowledge, such as classification and other particular applications [10], [11]. For example, Sun *et al.* [12] proposed a multi-scale spectral feature map fusion (MSFGF) band selection method for classification, which explored band selection from the perspective of multiple spatial scales. Unsupervised band selection uses general criteria, such as entropy. For instance, Yuan *et al.* proposed an unsupervised band selection method by dominant set extraction, which exploits the structurally meaningful information for the measures of band informativeness and independence [13]. Zhang *et al.* [14] proposed an immune clonal selection algorithm based on maximizing representativeness and minimizing redundancy (MRMR) to consider both representativeness and redundancy. Xu *et al.* [15] designed a similarity-based ranking (SR) strategy to evaluate the representativeness of each band by the product of average similarity and significant dissimilarity. Zhang *et al.* [16] proposed a marginalized graph self-representation (MGSR) method for band selection to explore the spatial information from diverse homogenous regions. In recent years, deep neural networks (DNNs) have been successfully applied in the field of HSIs. Xie *et al.* [17] proposed an unsupervised band selection method based on deep latent spectral representation learning, which adopts information entropy and correlation coefficient to assess the quality of selected bands. Mou *et al.* [18] proposed a deep reinforcement learning model for band selection in HSI classification, using information entropy and correlation coefficient to assess the quality of bands. Although such methods have achieved promising performance, there still exist several limitations. First, these methods are mainly based on single indicator to measure the bands, which will lead to insufficient band evaluation and difficulty in assessing the interactions between multiple criteria. Second, unsupervised band selection methods mainly use common criterion, without special consideration about the targets of interest. Therefore, unsupervised band selection which makes full use of the known targets and properties of HSI is more suitable for target detection.

Manuscript received April 23, 2022; accepted May 18, 2022. Date of publication May 23, 2022; date of current version June 3, 2022. This work was supported in part by the National Natural Science Foundation of China under Grant 61971082, Grant 61890964, and Grant 42101350; and in part by the Fundamental Research Funds for the Central Universities under Grant 3132019341 and Grant 3132022249. *(Corresponding author: Haoyang Yu.)*

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Digital Object Identifier 10.1109/TGRS.2022.3176856

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For the first problem, multiobjective optimization (MO) is considered to be a promising strategy, especially in dealing with multiple conflicting objectives for band selection. Gong *et al.* [19] proposed a multiobjective optimization band selection (MOBS) method based on the information entropy and the number of selected bands. Zhang *et al.* [20] proposed an MO-model-based framework, designing two objectives of band information and redundancy. Xu *et al.* [21] proposed an incorporated rank-based multiobjective band selection (IRMoBS) method, which does not have to set the band number manually. These algorithms mainly focus on optimizing the information and redundancy of the selected bands, which are not fully enough to evaluate the band set. In addition, the targets of interest are usually known for target detection. The existing multiobjective band selection methods mostly did not consider the target factor. Therefore, the selected bands are not suitable for target detection.

For MO, the optimal solution searching process is also important. Most band selection methods under the framework of multiobjective evolutionary algorithm are based on decomposition (MOEA/D) [19], [21]. On account of the problem that MOEA/D is hard to reach Pareto optimal, Pan *et al.* [22] proposed a boundary-intersection-based multiobjective band selection method (BiMOBS), reducing the sensitivity of the penalty factor and avoiding the emergence of weak Pareto optimal. Zhang *et al.* [20] optimized the multiobjective model through reorganization and mutation operations. It can be found that the evolutionary mechanism is well-suited to deal with the MO process, but it is also easy to fall into local optimization. Therefore, it is necessary to study an effective way to accomplish optimization.

As one of the major tasks of HSIs, target detection is to detect targets with known spectral characteristics from unknown background. There are many classical algorithms for target detection, including constrained energy minimization (CEM) [23], [24], adaptive cosine estimation (ACE) [25], adaptive matched filter (AMF) [26], etc. In recent years, researchers have done a lot of work to detect targets more effectively. For example, Chang and Chen [27] jointly combined data sphering and low-rank and sparse matrix decomposition (LRASMD) to perform orthogonal subspace projection (OSP) for target detection. Meerdink *et al.* [28] adopted multitarget multiple instance to estimate multiple-target signatures from training samples with imprecise labels, using ACE and spectral match filter (SMF) to optimize detection against a background. In addition, target detection algorithms based on band selection have also made corresponding progress. Wang *et al.* [29] proposed a constrained target band selection approach (CTBS) for multiple-target detection and sequential forward CTBS (SFCTBS) method. Xie *et al.* [17] adopted a band selection method based on deep latent spectral representation learning (DLSRL), which introduced a hierarchical optimization approach to achieve target detection with the selected subset. Although the above band selection methods specially designed for target detection can select representative band subsets for specific targets, they usually ignore the correlation between adjacent bands.

Aiming to overcome the insufficient evaluation of band subset, local optimal solution of multiobjective evolutionary algorithm, and target ignorance of solution definition, this article proposes an algorithm for HSIs, which is called target-oriented MO of band selection (TOMOBS). The main contributions of this article can be concluded as follows.

1) A new MO band selection model is proposed. This model simultaneously involves three factors, including information, noise, and redundancy, which can better describe the characteristics of the bands in hyperspectral. Swarm intelligence algorithm based on genetic theory is developed to optimize the MO band selection model, and the noninferior solution advantage matrix (NAM) is designed to avoid falling into local optimization, providing a quantitative and interpretable approach for the MO model.
2) Target-oriented evaluation mechanism is constructed to select the final solution from the Pareto front (PF), so the selected band subset is more conducive to target detection.

The remainder of this article is organized as the following. Related formulation and motivation are introduced in Section II. Then, Section III describes the proposed method. In Section IV, the experiments on two datasets, hyperspectral digital imagery collection experiment (HYDICE) and the airborne visible/infrared imaging spectrometer (AVIRIS), are implemented. Finally, Section V gives the conclusions.

II. Related Formulation and Motivation

In this section, we first introduce the concept of MO problem (MOP) and its solution. Then, the motivation for designing a new MO model is analyzed.

A. MOP and Solution

MOP is also called multi-criteria decision-making or vector optimization problem [30]. Considering the minimization f of m components, an MOP can be described as follows:

$$\min_{\text{subject to}\mathbf{x}\in\Omega} F(\mathbf{x}) = (f_1(\mathbf{x}), f_2(\mathbf{x}), \ldots, f_m(\mathbf{x}))^T \tag{1}$$

where $\mathbf{x}$ is the decision vector containing n decision variables, Ω is the decision space, $F(\mathbf{x}) \in R^m$ is the objective vector composed of m objective functions $f_i(\mathbf{x})$, and $i = 1, 2, \ldots, m$. Without loss of generality, the MO in this article is set to minimize.

1) Definition (Feasible Solution Set): The set composed of all feasible solutions in decision space Ω is called feasible solution set.

2) Definition (Dominated Relation in Objective Space): Dominated solution and non-dominated solution of MO minimization. Let $\mathbf{x}_a, \mathbf{x}_b \in \Omega$ be two feasible solutions of (1), called $\mathbf{x}_a$ dominating $\mathbf{x}_b$ and recorded as $\mathbf{x}_a \prec \mathbf{x}_b$, if and only if

$$\begin{cases} f_i(\mathbf{x}_a) \le f_i(\mathbf{x}_b), & \forall i \in 1, \ldots, m \\ f_j(\mathbf{x}_a) < f_j(\mathbf{x}_b), & \exists j \in 1, \ldots, m. \end{cases} \tag{2}$$

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For the two feasible solutions, $\mathbf{x}_a$ is called the non-dominated solution and $\mathbf{x}_b$ is the dominated solution. Usually, according to user preferences, a compromise solution is selected from the non-dominated solutions, i.e., the solution of MOP.

3) Definition (Pareto Optimal Solution): The non-dominated solution in the feasible solution set is called Pareto optimal solution, and Pareto optimal solution is also called non-inferior solution.

4) Definition (PF): The mapping of Pareto optimal solution in target space constitutes PF.

With the increase in the number of MOPs' objectives, the solution of the compromised state becomes more and more complex. In the process of solving MOPs, researchers found that swarm optimization algorithms can search in parallel and deal with a set of possible solutions simultaneously. When the components of a set of Pareto solutions cannot be improved at the same time, it can approach PF and solve multiobjective complicated problems [31]. In recent years, researchers have proposed many swarm intelligence optimization methods for MO, such as the nonintegrated sorting genetic algorithm II (NSGA-II) [32] and MOEA/D [33]. It is worth noting that the advantages of the swarm intelligence optimization method in solving MOPs make it also achieve impressive performance in solving HSI problems [34]–[37]. Here, this article improves a multiobjective swarm intelligence optimization algorithm. Section III shows the details of the proposed TOMOBS.

B. Motivation of Designing a New MO Model for Band Selection

The effective objective functions are very important to select the ideal band subset. As mentioned earlier, the existing algorithms mainly focus on optimizing the information and redundancy of the selected bands, which are not fully enough to evaluate the band set. Noise is inevitable in real HSIs and noisy bands often bring high information entropy, which affects the evaluation of the bands. Meanwhile, the key information preservation can better represent the original information of HSIs. Therefore, noise resisting and key information preserving are crucial for band selection.

Here in this section, the relationship between information entropy and signal-to-noise ratio (SNR) estimation of bands is analyzed. The SNR estimation and entropy of HYDICE data are shown in Fig. 1. It can be seen that some bands with high information entropy may have higher noise than the band with low entropy. On the other hand, the bands with similar information entropy may have different extent of noise. Furthermore, as shown in Fig. 1, the bands before number 50 mostly have high information entropy, but their noise is high too. To some extent, there is some conflict between information entropy and SNR estimation. To better represent the original information, exploring the tradeoff between these two aspects, this article first takes information entropy and SNR estimation as two objective functions, so that the band selection problem is transformed into an MOP.

As mentioned above, the two objective functions with a conflicting relationship are designed. However, it can be seen

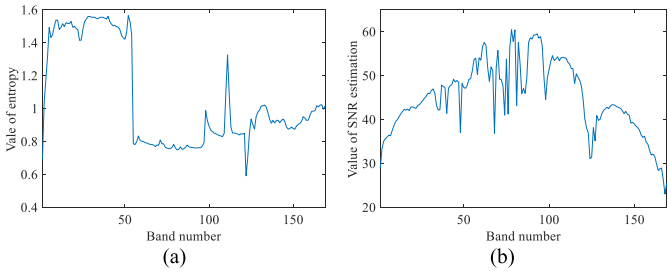


Fig. 1. (a) Entropy of each band on HYDICE. (b) SNR estimation of each band on HYDICE.

from HYDICE data that both high entropy and low entropy bands are mostly concentrated in a certain region of the bands, as shown in Fig. 1. If the band is selected according to the highest entropy value, multiple bands would be selected in a very small band section, which will result in strong redundancy. Therefore, it is better to select bands with relatively high entropy value in other band regions. Because of the mechanism of hyperspectral imaging equipment, the SNR is also high in some wavelength ranges and low in some ranges. In other words, the entropy and noise of adjacent bands generally have high correlation, which means huge redundancy among the adjacent bands. However, redundancy information could cause a large consumption of calculation and storage, which is not conducive to subsequent applications. Therefore, tradeoffs of these three aspects of information entropy, SNR estimation, and redundancy should be explored simultaneously. The proposed three objective functions in this article are detailed in Section III-A.

The calculation details of information entropy and SNR estimation are introduced below.

1) Information Entropy: According to the definition of information entropy proposed by Shannon [38], the quantification of information can be achieved. Therefore, for a random variable X, the definition of information entropy is as follows:

$$H(X) = -\sum_{x \in N} p(x) \log p(x). \tag{3}$$

In the formula, $p(x)$ represents the probability of occurrence of information $x \in N$. Correspondingly, HSIs can also be expressed by information entropy. The band consisting of $m \times n$ pixels is taken as a random variable X, where the probability of occurrence is calculated as the ratio of the band histogram to the number of pixels, which may be represented as follows:

$$p(x) = \frac{h(x)}{mn}. \tag{4}$$

Among them, $h(x)$ represents the grayscale histogram of the bands. Therefore, the information entropy can represent the information contained in the band. To select the band with a large amount of information, the information entropy is designed as one objective function.

2) SNR Estimation: Due to the noise pollution caused by the sensor imaging process, there are bands with high noise in the hyperspectral data, which reduces the reflectance difference between the target and the background. As described above, the band selection method based solely on the entropy value

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cannot guarantee the selection of a low noise band, which seriously affects the subsequent image interpretation. There is a conflict relationship between SNR estimation objective function and information entropy objective function. SNR estimation can determine the noise of the selected band. Therefore, SNR estimation of bands is used as another objective function in this article.

The noise level of HSIs varies from band to band, and some bands have more severe atmospheric absorption, making the signal unusable. The noise model is defined as follows:

$$Y = X + N(X). \tag{5}$$

Among them, $Y = (Y_1, Y_2, \ldots, Y_B)$ is the observed noisy data block, $X = (X_1, X_2, \ldots, X_B)$ is the data block, $N(X) = (n(X_1), n(X_2), \ldots, n(X_B))$ is the Gaussian white noise, and the variance of $n(X_i)$ is proportional to the average amplitude of band X_i.

Signal and noise can be separated by wavelet transform. After wavelet transform, the energy of the image is mainly concentrated in the low-frequency component, while the amplitude of the high-frequency component with small scale is small and the energy is low. Therefore, when the noise is large, the high-frequency component in the diagonal direction can be regarded as all noise. Because the independently identically distribution Gaussian white noise is still independently identically distribution Gaussian white noise after orthogonal wavelet transform, and the noise variance remains unchanged. Therefore, median absolute deviation (MAD) can be used to estimate the variance of noise in wavelet domain, that is,

$$\sigma_n = \text{MAD}/0.6745. \tag{6}$$

For image signal variance of $m \times n$ pixels is

$$\sigma_s = \frac{1}{m \times n} \sum_{i=1}^{m} \sum_{j=1}^{n} (X(i, j) - \mu)^2 \tag{7}$$

where μ is the mean value. According to the definition of SNR estimation

$$N = 10 \lg \frac{\sigma_s^2}{\sigma_n^2} \tag{8}$$

where N represents the SNR estimation of the selected band.

III. Proposed Method

In this section, we first describe the details of the proposed model and the main content of proposed method, and then provide the TOMOBS algorithm in detail.

A. New MO Band Selection Model

To better evaluate the bands of HSIs, an MO model is established. In this model, three objective functions are used to represent the information entropy, SNR estimation, and redundancy of the selected bands, respectively. The three objective functions are designed as follows:

$$\min F(\mathbf{x}) = \begin{cases} f_1(\mathbf{x}) = K \sum_{i=1}^{K} \dfrac{1}{H(\mathbf{x}_i)} \\ f_2(\mathbf{x}) = K \sum_{i=1}^{K} \dfrac{1}{N(\mathbf{x}_i)} \\ f_3(\mathbf{x}) = \dfrac{K(K-1)}{2} \sum_{i=1}^{K} \sum_{j=i+1}^{K} \dfrac{1}{D_{\text{JS}}(\mathbf{x}_i, \mathbf{x}_j)}. \end{cases} \tag{9}$$

Among them, K represents the number of bands in the selected band subset. $H(\mathbf{x}_i)$ is the information entropy of the ith selected band. $N(\mathbf{x}_i)$ represents the SNR estimation of the ith selected band. $D_{\text{JS}}(\mathbf{x}_i, \mathbf{x}_j)$ represents the JS divergence (JSD), which is used to measure the difference in the distribution between the ith band and jth band. If the average information entropy value is large, the amount of information contained in the selected band is large. If SNR estimation is large, the noise of the selected band is small. If the JSD value is large, the distribution difference between the two selected bands is large, and the redundancy included is small. These three objective functions need to be minimized at the same time. The calculation details of information entropy and SNR estimation are described in Section II-B.

1) Jensen–Shannon Divergence: From the definition of information entropy and SNR estimation objective functions, it can be seen that the band with large entropy and low noise is more likely to be selected, resulting in the problem of large redundancy between the selected adjacent bands.

In information theory, relative entropy is also called Kullback-Leibler (KL) divergence. It is an index that measures the difference between two probability distributions. Assuming that X is used to represent a random variable, the two probability distributions on the random variable are $p(x)$ and $q(x)$, and then KL divergence is defined as follows:

$$\begin{aligned} D_{\text{KL}}(p||q) &= \sum_{x \in X} p(x) \cdot \left(\log \frac{1}{q(x)} - \log \frac{1}{p(x)} \right) \\ &= \sum_{x \in X} p(x) \log \frac{p(x)}{q(x)}. \end{aligned} \tag{10}$$

The value range of D_{KL} is $[0, +\infty]$. However, it is impossible to accurately use D_{KL} to measure the difference between the two sets of data. Although D_{KL} is intuitively a metric or distance function, it is not a real metric or distance, because it has no symmetry, that is,

$$D_{\text{KL}}(p||q) \neq D_{\text{KL}}(q||p). \tag{11}$$

JSD [39] is a variant based on KL divergence, which solves the problem that KL divergence is asymmetric and does not satisfy the triangle inequality. For two probability distributions p and q, the JSD formula is as follows:

$$D_{\text{JS}}(p||q) = \frac{1}{2} \text{KL}\left(p||\frac{p+q}{2} \right) + \frac{1}{2} \text{KL}\left(q||\frac{p+q}{2} \right). \tag{12}$$

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From (12), JSD is the accumulation of two KLs. Expanding KL is as follows:

$$D_{\mathrm{JS}}(p||q) = \sum_{x\in X}\left(p(x)\cdot\log\frac{2p(x)}{p(x)+q(x)} + q(x)\cdot\log\frac{2q(x)}{p(x)+q(x)}\right). \tag{13}$$

The JSD is symmetric, and the range is [0,1], i.e., JSD measures the similarity between the sum of two probability distributions. When the two random distributions are the same, the JSD is 0, and completely different case is 1. The more similar, the smaller the JSD. JSD solves the asymmetric problem and satisfies symmetry

$$D_{\mathrm{JS}}(p||q) = D_{\mathrm{JS}}(q||p). \tag{14}$$

Compared with KL divergence, JSD is stable in calculation. It can determine the similarity of two probability distributions regardless of symmetry or value range more accurately. It is widely used in different fields, but has not been used in the field of hyperspectral band selection.

To prevent selecting redundant bands, we use JSD to quantify the change in band distribution. According to (13), the expression for the JSD of the ith band and jth band can be obtained as follows:

$$\begin{aligned} D_{\mathrm{JS}}(\mathbf{x}_i||\mathbf{x}_j) &= \frac{1}{2}\left(\sum_{y\in\Omega} p_i(y)\log\frac{p_i(y)}{\frac{1}{2}p_i(y)+\frac{1}{2}p_j(y)}\right) \\ &\quad + \frac{1}{2}\left(\sum_{y\in\Omega} p_j(y)\log\frac{p_j(y)}{\frac{1}{2}p_i(y)+\frac{1}{2}p_j(y)}\right) \end{aligned} \tag{15}$$

where random variables $\mathbf{x}_i$ and $\mathbf{x}_j$ are set in Ω space, $p_i(y)$ and $p_j(y)$ are the probability distributions of random variable. According to the definition of JSD, the JSD value is large, the distribution difference between the selected two bands is large, and the redundancy contained is small. The JSD is more accurate in judging the similarity of the selected bands. Therefore, the JSD criterion is designed as the last objective function.

In general, the main reasons for choosing the three objective functions are: 1) to retain of critical information; 2) to avoid selecting high noise bands; and 3) to remove of redundant information.

B. Definition of NAM

In response to the requirements of MO method, this article proposes a NAM method, which can clearly describe the relationship within the population vector space and provide an accurate guide. At the same time, the specific advantages of the selected global optimal individuals in each dimension are described. The global best individual (gbest) selection based on the NAM is one of the key improvements to the swarm intelligence optimization algorithm.

According to the definition of Pareto dominated relation, for $f_i(\mathbf{x}), i = 1 \sim n$ with n objective components, arbitrarily given two m-dimensional decision vectors $\mathbf{a} = (a_1, a_2, \ldots, a_m)^T$ and $\mathbf{b} = (b_1, b_2, \ldots, b_m)^T$, if the following two conditions are true, it is called $\mathbf{a}$ dominate $\mathbf{b}$, where $\mathbf{a}$ is the non-dominated solution.

1) $\forall i \in \{1, \ldots, n\}$, $f_i(\mathbf{a}) \le f_i(\mathbf{b})$
2) $\exists i \in \{1, \ldots, n\}$, $f_i(\mathbf{a}) < f_i(\mathbf{b})$.

The non-dominated solution can be selected from the definition of the dominant relationship. At the same time, the numerical relationship between the non-dominated solutions $\mathbf{x}^i = (x_1, x_2, \ldots, x_m)^T$ and $\mathbf{y}^i = (y_1, y_2, \ldots, y_m)^T$, $i \in [1, 2, \ldots, n]$ can also be analyzed from the perspective of m-dimension. Therefore, to reflect the detailed numerical relationship between individuals, this article defines an advantage concept: when an individual is smaller than another individual in a certain dimension, it indicates that this individual dominates other individuals in this dimension, and this relationship is set to 0. Otherwise, the advantage is used to indicate the difference between the two, as defined in the following equation:

$$A(\mathbf{x}^i, \mathbf{y}^i) = \begin{cases} \mathbf{x}^i - \mathbf{y}^i, & \text{if } \mathbf{x}^i \ge \mathbf{y}^i \\ 0, & \text{otherwise} \end{cases} \tag{16}$$

where $i \in [1, 2, \ldots, n]$. According to (16), we can see the relationship between the two individuals $\mathbf{x}^i = (x_1, x_2, \ldots, x_m)^T$ and $\mathbf{y}^i = (y_1, y_2, \ldots, y_m)^T$ in m-dimension. If this relationship is extended to the whole space, the NAM can accurately describe the relationship in the vector space of the population. The following is an example to illustrate the calculation process of NAM.

To avoid excessive gap between individual indicators to affect the choice of individuals, the min-max normalization method is adopted in this article. According to the number of objectives in this article, the individual can be expressed as $\mathbf{x} = (x_1, x_2, x_3)^T$, and the normalization of individual objective value can be expressed as

$$f_{\mathrm{norm}}(\mathbf{x}) = \frac{f(\mathbf{x}) - \min(P_m)}{\max(P_m) - \min(P_m)}, \quad m = 1, 2, 3 \tag{17}$$

where $\min(P_m)$ is the minimum value of each objective in the population, and $\max(P_m)$ is the maximum value of each objective in the population.

To simplify the operation, only five individuals are selected for description, i.e., $n = 5$. Since this article uses three objective functions, each individual is a 3-D vector, $m = 3$, $\mathbf{a} = (a_1, a_2, a_3)^T$, $\mathbf{b} = (b_1, b_2, b_3)^T$, $\mathbf{c} = (c_1, c_2, c_3)^T$, $\mathbf{d} = (d_1, d_2, d_3)^T$, and $\mathbf{e} = (e_1, e_2, e_3)^T$. According to (16), calculate NAM of five individuals, as shown in the following equation (18), as shown at the bottom of the next page.

For the NAM, each column represents the dominance degree of the corresponding individuals in the target space. Therefore, this article can use the 1-norm of matrix NAM to further analyze the relationship between individuals and obtain their distinguishability. For the 1-norm of matrix NAM, as shown in following equation:

$$\|\mathrm{NAM}\|_1 = \max_{1\le j\le n}\left(\sum_{i=1}^{n}|A_{ij}|\right). \tag{19}$$

By analyzing the maximum absolute column and norm, we can get the accumulation of the difference in each column.

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The maximum column corresponding to the norm value is the global best individual, and then the solution with Pareto optimality is obtained. This relationship transformation from target space to NAM space has two advantages. On one hand, the gbest in the population can be selected to avoid falling into local optimization, so that individuals in the population converge to the optimal solution in the form of the population by following the best leader. On the other hand, the solution of multiobjective space is discriminative, which contributes to the scalability of the solution process of MOPs.

C. Target-Oriented Evaluation Mechanism

According to different application requirements, the optimal solution selection ways of PF are different. At present, the optimal solution is usually selected by trial-and-error [20], decision-maker (DM) [40], clustering [41], and knee-point-based selection method [42]. Most of these methods are based on classification performance, so the selected band subset is suitable for classification task, but not for target detection task. With respect to target detection, known target spectrums are given. The current researches on the method of band selection for target detection show that the combination of target characteristics in the process of band selection is better than that without consideration [17], [29]. This algorithm is aimed at the band selection of target detection, so it is more reasonable to consider the influence of target.

Specifically, to reflect the trend degree of band subsets to the spectral characteristics of known targets, a target-oriented evaluation mechanism is designed for optimal result selection in this article. The evaluation mechanism adopts the mean method, which can be defined as follows:

$$M = \left| \frac{1}{d} \sum_{i=1}^{d} \text{mean}(D_i) - \frac{1}{K} \sum_{i=1}^{K} \text{mean}(B_i) \right| \tag{20}$$

where D_i represents the spectral characteristic value of the ith known target, and d is the number of known targets. B_i represents the characteristic value of the ith selected band, and K is the number of selected bands. The smaller the difference M, the stronger the representation ability of the selected band subset to the target. The band subset with the smallest difference is the final result selected from the PF in this article. Through the operation of the target-oriented evaluation mechanism, it can be seen that TOMOBS is a band selection method specially designed for target detection.

D. TOMOBS Algorithm and Solution

It is challenging to establish a balance between exploration and utilization in the multiobjective swarm optimization method. To resolve the problem, this work conducts two types of research. On one hand, to study the global parameter space, this article proposes the NAM technique to select global best individual, which achieves the balance between population diversity protection and rapid convergence. On the other hand, to refine the local optimal search, this article proposes a swarm intelligence algorithm based on the genetic mechanism to implement adaptive crossover and mutation. These operations decrease linearly with the number of iterations, high-quality individuals have more opportunities to be copied, and low-quality individuals are removed from the population to further reduce the individual search space. After iteration, the quality of the band subset is improved through the above operations. Focusing the search on more effective areas, the TOMOBS algorithm can obtain better solutions in a shorter time. Under the target-oriented evaluation mechanism, the final result will be obtained from the PF. The framework of the proposed method is shown in Fig. 2.

Different from the real number coding method of generating initial individual in random mode, this article uses a coding method combining uniform randomization. First, the dataset is divided into several subintervals with the same length as the number of selected bands, and each subinterval after uniform division has only one chance to generate coding values. In this way, the uniformity of coding value distribution in decision space is guaranteed to the greatest extent. Then random values are generated in the selected subinterval as real coded values. This uniform and random real number coding method increases the diversity of genes and makes individuals contain more information, which is beneficial for the algorithm to search the optimal solution in a larger range, and provides a good start for the iterative search of the TOMOBS method. Individual $b = (b_1, b_2, \ldots, b_k)$ is the encoding of variable $\mathbf{x}$ in (1), where b_k is the kth selected band.

To effectively detect the target, this article adopts the virtual dimensionality (VD) method [43], [44] to determine the number of bands without supervision. VD is a method for determining the number of bands based on the different features of the spectrum in hyperspectral data under the Neyman–Pearson detection theory. It can be seen that for different data, the number of selected bands n_{BS} by the VD method will change with the target signatures. Because hyperspectral data record a scene from the observed area with a number of spectral signatures. For the development of hyperspectral data, the VD method is effective and suitable. Therefore, this article uses the VD method to determine the number of selected bands.

$$\text{NAM} = \begin{bmatrix} 0, & a_1-b_1+a_2-b_2+a_3-b_3, & a_1-c_1+a_2-c_2+a_3-c_3, & a_1-d_1+a_2-d_2+a_3-d_3, & a_1-e_1+a_2-e_2+a_3-e_3 \\ b_1-a_1+b_2-a_2+b_3-a_3, & 0, & b_1-c_1+b_2-c_2+b_3-c_3, & b_1-d_1+b_2-d_2+b_3-d_3, & b_1-e_1+b_2-e_2+b_3-e_3 \\ c_1-a_1+c_2-a_2+c_3-a_3, & c_1-b_1+c_2-b_2+c_3-b_3, & 0, & c_1-d_1+c_2-d_2+c_3-d_3, & c_1-e_1+c_2-e_2+c_3-e_3 \\ d_1-a_1+d_2-a_2+d_3-a_3, & d_1-b_1+d_2-b_2+d_3-b_3, & d_1-c_1+d_2-c_2+d_3-c_3, & 0, & d_1-e_1+d_2-e_2+d_3-e_3 \\ e_1-a_1+e_2-a_2+e_3-a_3, & e_1-b_1+e_2-b_2+e_3-b_3, & e_1-c_1+e_2-c_2+e_3-c_3, & e_1-d_1+e_2-d_2+e_3-d_3, & 0 \end{bmatrix} \tag{18}$$

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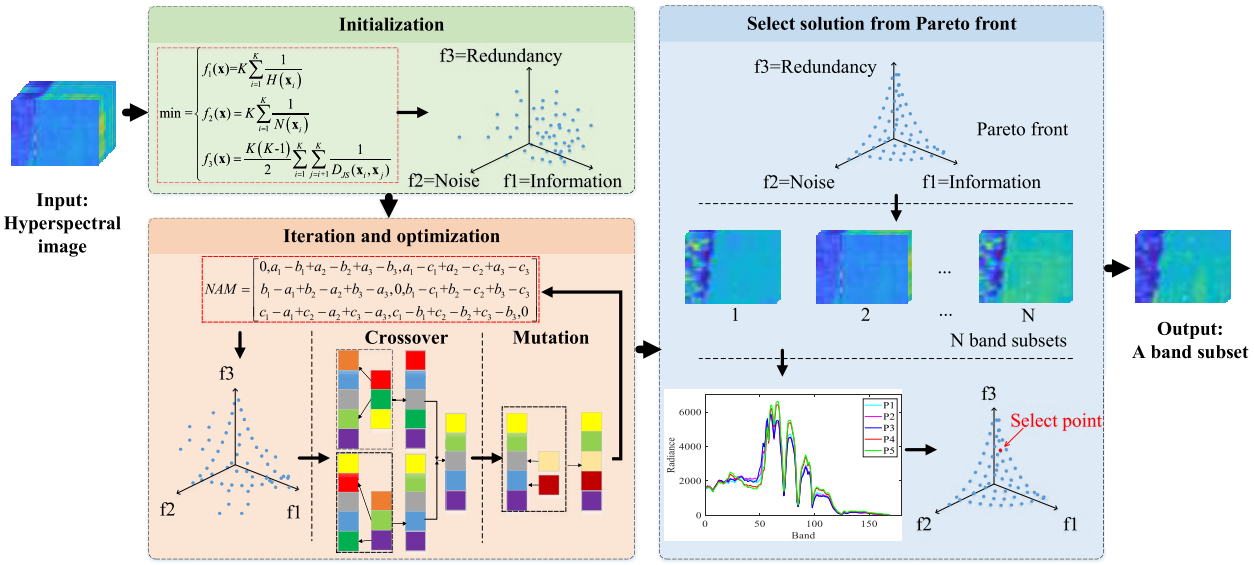


Fig. 2. Framework of the proposed method.

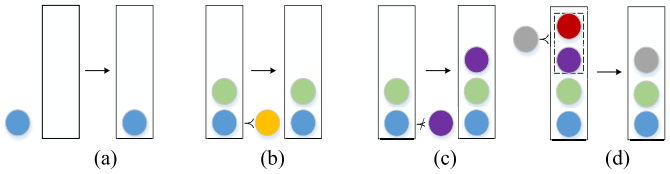


Fig. 3. Rep set storage strategy. (a) Rep set is empty. (b) Newly selected solution is dominated by any solution in the Rep set. (c) Newly selected solution is not dominated by the Rep set solution. (d) New solution dominates some Rep set members.

The detailed process of the TOMOBS algorithm is as follows.

1) Initialization:

1) Use the VD method to determine the number of selected bands. Set the initial parameters, such as the number of iterations, the number of populations, and algorithm parameters.
2) Initialize the population P_t. Calculate the values of the proposed objective functions of all individuals in the population P_t, and determine the noninferior solution Rep set. The Rep set needs to meet the following requirements.
   If the Rep set is empty, any new non-dominated solutions are directly put into the Rep set, as shown in Fig. 3(a). When there are members in the Rep set, the newly selected solution needs to be compared with the solution in the Rep set. If the newly selected solution is dominated by any solution in the Rep set, it cannot be added to the Rep set, as shown in Fig. 3(b). If the newly selected solution is not dominated by the Rep set solution, it can be added to the Rep set, as shown in Fig. 3(c). If a new solution dominates some Rep set members, the new solution is accepted, and those Rep set members dominated by the new solution are automatically discarded, as shown in Fig. 3(d).
3) The initial population P_t is generated, and the noninferior solution in P_t is copied to the Rep set. Then, the coding method combined with uniform randomization is adopted for individuals. Calculate the density information of individuals in Rep set, divide the target space into small areas with grids, and take the number of individuals contained in each area as the density information of individuals. The more individuals the mesh contains, the greater the density value, and vice versa. The current number of iterations is t. When t is less than the maximum number of iterations MaxIt, continue the contents of 2)–4).

2) Evolution Produces the Next Generation of Population: Let individual i of the current generation complete the contents of 1) to 2) when i is less than the population size.

1) According to the global optimal probability p, select an individual in Rep set as the gbest. The combination of roulette operator random selection and NAM is used to select gbest to avoid falling into local optimization. Specifically, when the random number is less than the probability of global optimal selection p, the NAM in Rep set is calculated, and the maximum individual corresponding to 1-norm is selected as gbest. The gbest obtained from the NAM of the noninferior solution ensures that the individuals in the initial population move to the highest quality region of the search space. If the values of the maximum absolute column and norm correspond to more than one column, the individuals with higher density congestion values are selected. When the random number is greater than p, the roulette operator is used to randomly select a nondominated solution in Rep set as gbest, and the random selection can increase the diversity of individuals.
2) Update the position and velocity of individuals in the population, and search for the optimal solution under the guidance of gbest and pbest

$$v_i = \omega v_i + c_1 r_1 (p_i - x_i) + c_2 r_2 \left(p_g - x_i\right) \quad (21)$$

$$x_i = x_i + v_i \quad (22)$$

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where v_i represents the velocity of the current individual position x_i. p_i represents the vector of the current individual best index, and p_g represents the vector of the current global best index. c_1 and c_2 are the learning factors that regulate individuals, respectively. $\omega > 0$ is the inertia weight. r_1, r_2 are the pseudorandom number uniformly distributed in the range of [0, 1].

3) Genetic Operation of Crossover and Mutation: According to the adaptive crossover rate and mutation rate, the genetic operation of crossover and mutation is carried out for the new population.

a) Crossover operation: Calculate the crossover probability according to the current number of iterations

$$pc = \frac{1-(t-1)^{\frac{1}{\mathrm{cu}}}}{\mathrm{MaxIt}-1} \tag{23}$$

where t is the current number of iterations, MaxIt is the maximum number of iterations, and cu is the crossover coefficient. As the number of iterations increases, the crossover rate decreases gradually. When the random number is less than the crossover probability, the product of crossover rate and individual population is calculated to obtain the number of crossover pool. Randomly select two individuals from the crossover pool, cross them, and update the position, velocity, and band subset of the offspring

$$\mathrm{child}x_i = \theta x_1 + (1-\theta)x_2 \tag{24}$$

where $\mathrm{child}x_i$ represents the position of the ith new child, θ is a random number, x_1 represents the position of parent A, and x_2 represents the position of parent B

$$\mathrm{child}v_i = \frac{(v_1+v_2)*\|v_1\|}{\|v_1\|+\|v_2\|} \tag{25}$$

where $\mathrm{child}v_i$ represents the speed of the ith new generation, v_1 represents the position of parent A, and v_2 represents the position of parent B.

The new solution is generated by the following steps: first, choose two band subsets A and B to serve as parents. $B\backslash A$ and $A\backslash B$ may be produced from the bands present in subsets A and B. Select h element in $B\backslash A$ to replace elements in A, and h elements in B are replaced by elements in $A\backslash B$ at the same time. The element h is found by comparing the smallest number of elements in $B\backslash A$ and $A\backslash B$. $A1$ and $B1$ indicate the new combination following replacement, respectively. Equation (26) is used to compare the values of $A1$ and $B1$ to identify a subset that is favorable to band selection. The solution with a large amount of information is selected as the final new solution C

$$f(\mathbf{x}) = K \Big/ \sum_{i=1}^{K} H(\mathbf{x}_i). \tag{26}$$

As illustrated in Fig. 4, parent subsets A and B include the same elements but are not precisely the same. In comparison, the values $B\backslash A$ and $A\backslash B$ are obtained. According to the size of the two, three bands of A and B are chosen to replace the bands at random places, yielding new solutions $A1$ and $B1$. Calculate the amount of information in $A1$ and $B1$, and select the greater of the two as the new solution C.

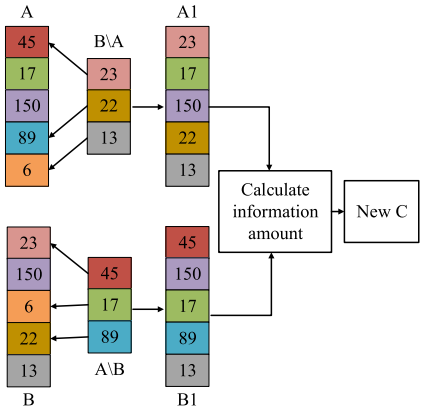


Fig. 4. Schematic of genetic operation.

b) Mutation operation: Calculate the mutation probability according to the current number of iterations

$$pm = \frac{(1-(t-1))^{\frac{1}{\mathrm{mu}}}}{\mathrm{MaxIt}-1} \tag{27}$$

where mu is the coefficient of mutation. With the increase in the number of iterations, the variation rate decreases gradually. When the random number is less than the mutation probability, it is updated according to the optimal position and velocity of the current individual, and a new band subset is randomly generated to prevent the individual from falling into the local optimal solution.

4) Update Rep Set: After the new generation population P_{t+1} is evolved, calculate the values of three objective functions of all individuals in the population P_{t+1}. The noninferior solution in P_{t+1} is saved in Rep set. When the number of individuals in the Rep set exceeds the specified size, redundant individuals need to be deleted to maintain a stable Rep set size. When t reaches MaxIt, go to 5) to continue. Otherwise, return to 2). The number of iterations is $t = t + 1$ until the iteration meets the threshold requirements.

5) Get N Band Subsets: According to the individual information in Rep set, N groups of band subsets are obtained.

6) Output A Band Subset: Using the proposed target-oriented evaluation mechanism, a subset of the N band subsets is selected as the final result.

From the previous description, key information, low noise, and low redundancy band subsets participate in the optimization process. In addition, the objective function based on information entropy can select bands with more key information. Combined with the objective function based on SNR estimation, the high noise bands can be well-reduced. At the same time, the objective function based on JSD can reduce the redundancy between bands. Therefore, it is inferred that the TOMOBS algorithm can obtain a subset of bands with strong representational capability. A detailed step-by-step implementation of TOMOBS is given as follows.

IV. Experimental Analysis

This article evaluates the proposed band selection algorithm by target detection. In this part, full band is a baseline to adopt all spectral bands for experiment. The following two MO band

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Algorithm 1 Algorithm TOMOBS

Input: Hyperspectral image.

Output: A band subset.

Step 1. Initialization: Determine the number of selected bands n_{BS} by VD. Randomly initialize population P_t and initialize iteration number t. Calculate the three objective functions values of all individuals in the population P_t by (9). Get noninferior population Rep set.

Step 2. Update noninferior population: Calculate NAM and information of individuals in Rep and generate global best individual p_g and individual best p_i. Update individual velocity and position by (21) and (22).

Step 3. Genetic operation of crossover and mutation: Implement crossover and mutation on Rep to get updated population P_{t+1}. Calculate the three objective functions values in the population P_{t+1} by (9) and update Rep.

Step 4. If t reaches the max number of iterations MaxIt, go to step 5. Otherwise, go to step 2, $t \leftarrow t + 1$.

Step 5. Get multiple band subsets in Rep.

Step 6. Use target-oriented evaluation mechanism and select a band subset from multiple band subsets.

selection algorithms are used as comparison methods. The first method is the MOBS algorithm [19], which is a classical multiobjective band selection method to obtain an appropriate number of band subsets through the MOEA method. The second method is the BiMOBS algorithm [22], which avoids weak Pareto solutions and the same as the objective function of MOBS. Meanwhile, three unsupervised band selection algorithms are used as comparison algorithms. The third method is the fast neighborhood grouping (FNGBS) [45], which uses local density and information entropy to obtain the most relevant and informative bands. The MGSR method is adopted as the fourth comparison algorithm [16], because MGSR can well explore the difference between diverse homogeneous regions. The fifth method is the SFCTBS [29] method, which minimizes the variance resulting from the background (BKG) for target detection. The main reason for comparison is that MGSR, FNGBS, and SFCTBS select unlike bands for different targets, which is consistent with this article. For the fairness of comparison, the typical target detectors, ACE and CEM, are used to detect the targets of the band subset generated by the proposed method and the comparison algorithms in this article. The CEM and ACE are used for target detection because they only need the knowledge of detecting targets but not anything else. We analyze the detection effect of CEM and ACE. In the following experiment, we set the number of iterations to 50, the initial population size as 100, and the maximum value of Rep is 50.

To evaluate detection performance, this article adopts 3-D receiver operating characteristic (ROC) of detection result to quantify the effect of the selected bands, where the algorithms can be analyzed from different views by three types of 2-D ROC curves. These three types of 2-D ROC curves include 2-D ROC curve of (P_D, P_F), 2-D ROC curve of (P_D, τ), and 2-D ROC curve of (P_F, τ). The area under curve (AUC) of (P_D, P_F) value is the area under the curve of the ROC curve enclosed by the coordinate axis. Notably, the AUC of (P_F, τ) is used to evaluate CEM and ACE in BKG suppression. The better the detection effect, the higher the AUC of (P_D, P_F) and AUC of (P_D, τ) value. The smaller the AUC of (P_F, τ), the lower the probability of false detection, which means the better the detection effect.

TABLE I

n_{BS} AND $2n_{\mathrm{BS}}$ ESTIMATED FOR HYDICE DATA BY VD

Data Set	n_{BS}	$2n_{BS}$
HYDICE	9	18

The probability detection rate P_D and false alarm rate P_F are defined as follows:

$$P_D = \frac{N_d}{N_t} \tag{28}$$

$$P_F = \frac{N_f}{N}. \tag{29}$$

Among them, N_d represents the number of target pixels detected under a given threshold τ, N_t represents the number of real target pixels in the image, N_f represents the number of background pixels that are misjudged as a target, N represents the total number of pixels in the image, and the threshold τ for calculating P_D and P_F.

Although there are many real HSI datasets that can be used in experiments, such as Indian Pines, Salinas, and University of Pavia, these datasets are widely used in HSI classification rather than target detection. Therefore, experiments are performed on two real HSI datasets widely used in target detection to evaluate the performance of the proposed algorithm and comparison methods. Two data results are described in detail below.

A. HYDICE Data Results

The first set of data is an airborne HYDICE scene, as shown in Fig. 5(a). This scene has been studied extensively by many hyperspectral target detection reports [29], [46], [47]. HYDICE contains 169 spectral bands with size of 64 × 64. There contains 15 square panels, with three different sizes, 3 × 3, 2 × 2, and 1 × 1 m^2, respectively. The ground truth of 19 red (R) panel pixels is shown in Fig. 5(b), with the pixels in yellow (Y pixels) indicating panel pixels mixed with the background. From Fig. 5(b), there are five-panel spectral signatures $\mathbf{p}_1$–$\mathbf{p}_5$, where $\mathbf{p}_i$ is generated by the center pixel of the red panel in row i. Moreover, $\mathbf{p}_2$ and $\mathbf{p}_3$ panels are the targets of different colors with the same material. The last two lines are also the same. TOMOBS uses these panel signatures as desired target signatures to reflect the panels in each row. The spectra of $\mathbf{p}_1$–$\mathbf{p}_5$ are shown in Fig. 5(c).

As said above, to effectively detect the target, this article adopts the VD method in the selection of the number of bands. For the HYDICE dataset, Table I lists the n_{BS} and $2n_{\mathrm{BS}}$ values of VD with a false alarm rate of 10^{-4}.

The proposed method and the five-contrast band selection methods of selected band subsets with 18 bands on HYDICE are shown in Table II. It can be seen that SFCTBS, MGSR,

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TABLE II
18 BAND SUBSETS SELECTED BY VARIOUS METHODS ON HYDICE

	Method	Band Subset
	Full bands	1:1:169
	MOBS	98,124,13,167,28,26,105,146,145,78,148,143,106,112,35,67,110,48
	BiMOBS	96,49,93,128,144,15,10,151,43,86,89,145,38,76,92,160,141,79
$\mathbf{p}_1$	SFCTBS	169,127,122,105,112,149,130,116,128,123,148,17,22,2,106,109,142,132
$\mathbf{p}_1$	MGSR	60,61,66,59,65,67,57,62,56,68,64,78,77,76,79,63,58,80
$\mathbf{p}_1$	FNGBS	2,13,18,30,43,46,61,67,79,96,100,105,112,126,130,143,155,160
$\mathbf{p}_1$	TOMOBS	3,17,28,128,64,122,57,162,106,97,90,22,112,33,133,140,150,157
$\mathbf{p}_2$	SFCTBS	122,128,133,154,143,137,142,148,153,11,27,19,6,138,46,49,42,129
$\mathbf{p}_2$	MGSR	60,61,59,66,57,65,67,56,64,62,68,78,77,76,79,63,58,55
$\mathbf{p}_2$	FNGBS	3,15,18,30,43,49,61,67,76,89,99,107,112,126,130,135,155,160
$\mathbf{p}_2$	TOMOBS	143,6,19,26,30,121,41,57,68,129,153,76,82,96,100,111,132,146
$\mathbf{p}_3$	SFCTBS	122,126,53,41,118,111,15,52,45,51,38,107,44,5,49,26,37,98
$\mathbf{p}_3$	MGSR	60,61,57,59,66,67,65,56,62,68,64,78,63,77,76,79,55,58
$\mathbf{p}_3$	FNGBS	2,13,18,30,43,49,59,65,77,94,100,107,111,122,130,141,155,163
$\mathbf{p}_3$	TOMOBS	136,96,27,33,140,52,90,70,82,68,146,105,54,127,37,159,15,46
$\mathbf{p}_4$	SFCTBS	123,166,50,128,158,101,152,141,149,46,153,25,134,9,38,112,104,140
$\mathbf{p}_4$	MGSR	66,65,67,60,61,68,64,59,62,78,77,76,79,63,57,80,75,69
$\mathbf{p}_4$	FNGBS	3,13,18,30,43,49,60,66,76,89,100,105,110,126,130,139,155,162
$\mathbf{p}_4$	TOMOBS	99,50,26,32,141,54,60,71,110,84,105,100,111,114,126,139,147,156
$\mathbf{p}_5$	SFCTBS	122,49,25,17,39,154,136,28,153,18,27,38,52,132,155,26,151,14
$\mathbf{p}_5$	MGSR	66,67,65,60,61,68,64,59,62,78,77,76,79,63,57,80,75,69
$\mathbf{p}_5$	FNGBS	4,15,18,30,42,45,61,67,78,95,99,107,111,122,130,139,155,160
$\mathbf{p}_5$	TOMOBS	132,113,123,25,101,136,99,124,14,38,115,165,28,151,52,90,109,119

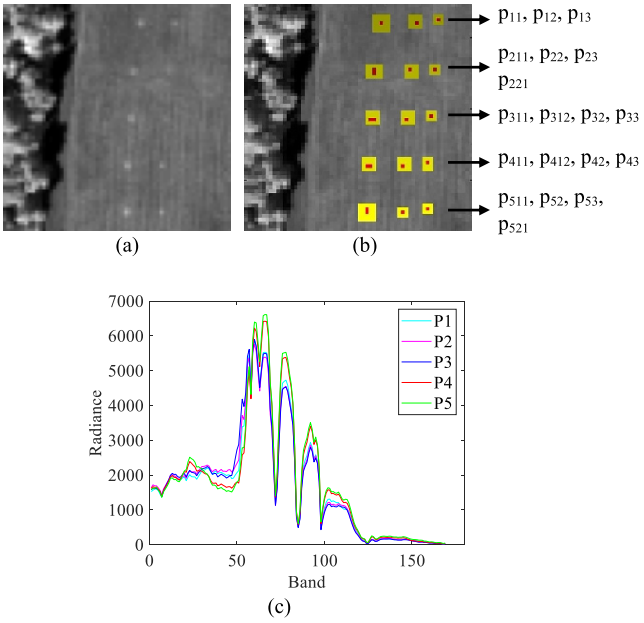


Fig. 5. (a) HYDICE scene. (b) Ground-truth map of the 15 panels. (c) Spectra of $\mathbf{p}_1$–$\mathbf{p}_5$.

FNGBS, and TOMOBS choose different band subsets for different targets. More specifically, SFCTBS is a band method for detection of multiple targets. MGSR can explore the difference between diverse homogeneous regions. FNGBS can mine the context information of the whole spectral bands. Due to the target-oriented evaluation mechanism, there are different selected band sets for different target signatures in TOMOBS. Theoretically, these four methods may have better effect on target detection.

When the number of bands in Table I is $2n_{\mathrm{BS}}$, Fig. 6 shows the PF of the selected band set with 18 bands on HYDICE, which is obtained by considering the three objective functions

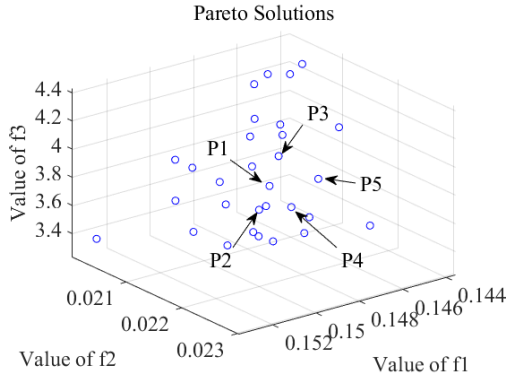


Fig. 6. Three-dimensional PF of the selected band set and TOMOBS method's target selection results for 18 bands on HYDICE.

simultaneously. As mentioned before, these points represent the optimal solution of the TOMOBS method, and each point represents a subset of 18 bands.

As shown in Fig. 6, the optimal point selected by the $\mathbf{p}_1$, $\mathbf{p}_3$, and $\mathbf{p}_5$ targets are far away from the horizontal axis, which means that the high information entropy and low noise effect of the $\mathbf{p}_1$, $\mathbf{p}_3$, and $\mathbf{p}_5$ targets play a more important role in detection. The point selected by the $\mathbf{p}_2$ target is closer to the horizontal axis, which means that $\mathbf{p}_2$ target reduction of redundancy is more important for detection than high information entropy and low noise effect. The point selected by $\mathbf{p}_4$ target is in the middle, which means that the effect of the three objective functions on $\mathbf{p}_4$ target is equally important. This is consistent with the facts that $\mathbf{p}_2$ is in the same material with $\mathbf{p}_3$, and $\mathbf{p}_4$ is in the same material with $\mathbf{p}_5$, making them difficult to be separated by common criteria but the divergence of adjacent spectra. The relationship between the three optimization objectives is detailed in Section IV-C.

The CEM and ACE detection results using 18 bands by the proposed method and the five-contrast band selection methods

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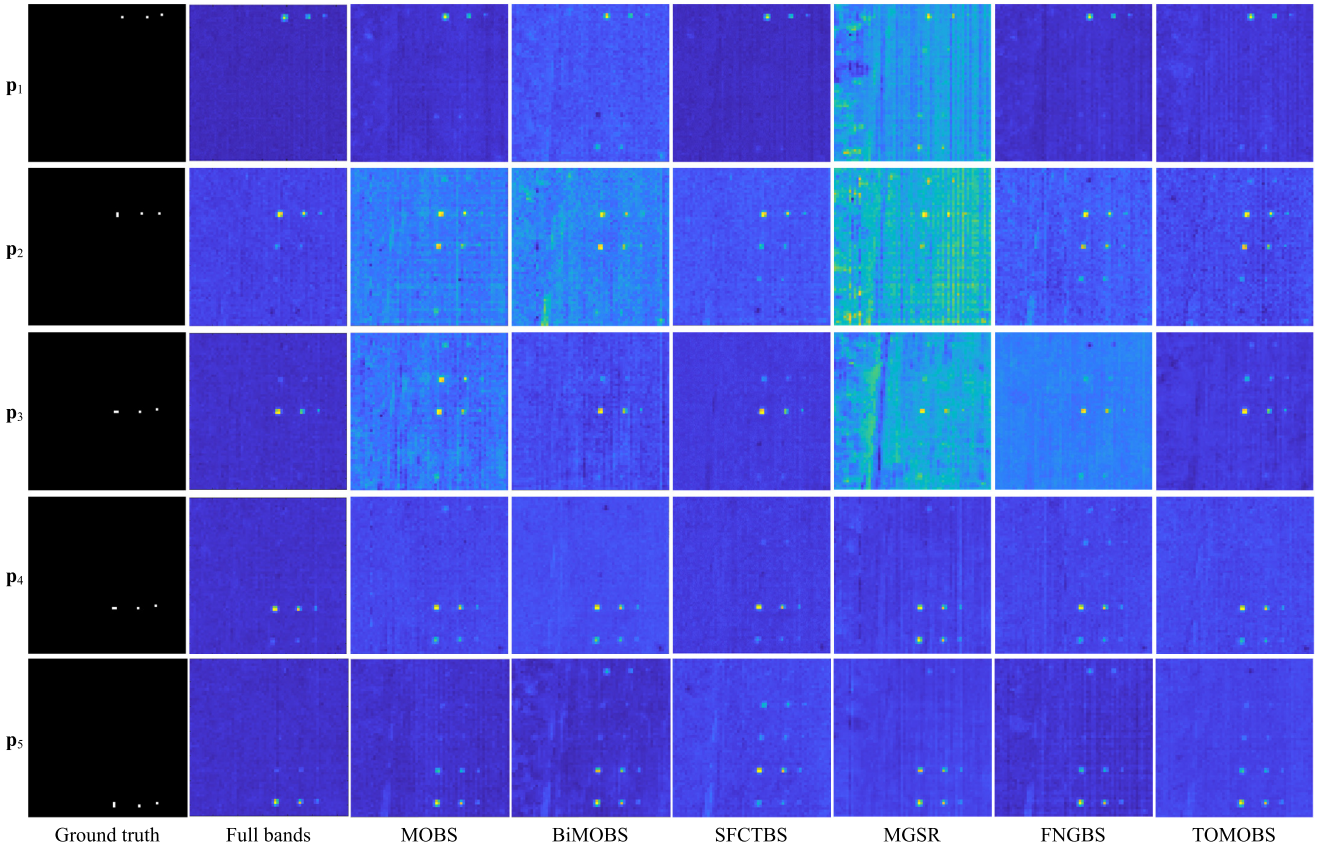


Fig. 7. CEM detection of five row panels using 18 bands selected by the six band selection algorithms and full bands in Table II.

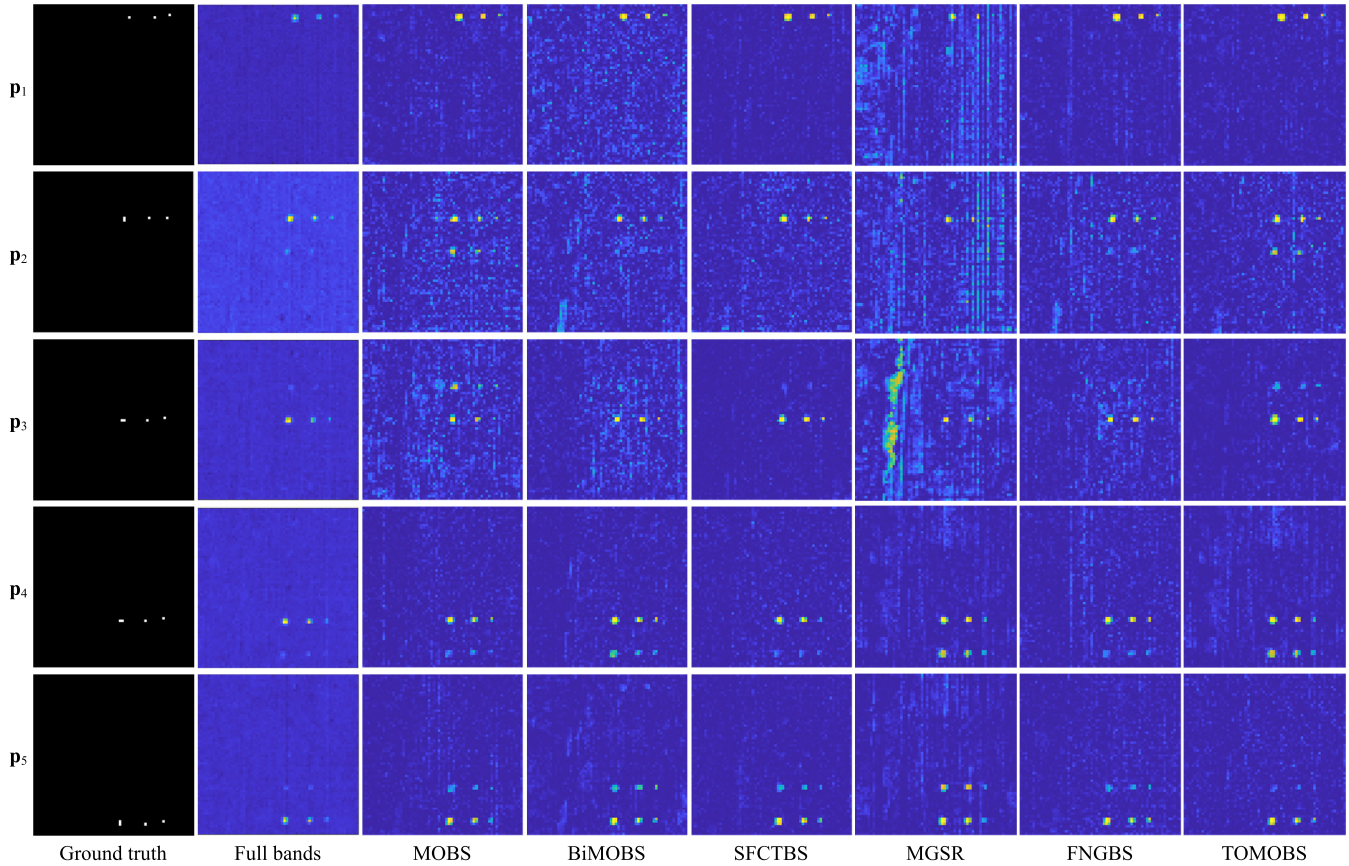


Fig. 8. ACE detection of five row panels using 18 bands selected by the six band selection algorithms and full bands in Table II.

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TABLE III

AUC OF (P_D, P_F) VALUES OF THE BAND SUBSET WITH 18 BANDS OF BAND SELECTION METHODS AND FULL BANDS FOR FIVE ROW TARGETS ON HYDICE BY CEM DETECTION

Method	$\mathbf{p}_1$	$\mathbf{p}_2$	$\mathbf{p}_3$	$\mathbf{p}_4$	$\mathbf{p}_5$	Average
Full bands	**0.9993**	**0.9998**	0.9997	**0.9998**	**0.9998**	**0.9996**
MOBS	0.9990	0.9991	0.9983	0.9995	0.9987	0.9989
BiMOBS	0.9984	0.9971	**0.9998**	0.9993	0.9985	0.9986
SFCTBS	**0.9992**	**0.9995**	**0.9996**	**0.9997**	**0.9995**	**0.9995**
MGSR	0.9949	0.9987	0.9994	0.9991	0.9987	0.9981
FNGBS	0.9991	0.9991	**0.9996**	0.9993	0.9991	0.9992
TOMOBS	**0.9992**	**0.9995**	**0.9996**	**0.9997**	**0.9995**	**0.9995**

TABLE IV

AUC OF (P_D, τ) VALUES OF THE BAND SUBSET WITH 18 BANDS OF BAND SELECTION METHODS AND FULL BANDS FOR FIVE ROW TARGETS ON HYDICE BY CEM DETECTION

Method	$\mathbf{p}_1$	$\mathbf{p}_2$	$\mathbf{p}_3$	$\mathbf{p}_4$	$\mathbf{p}_5$	Average
Full bands	**0.6033**	0.8387	0.7437	0.7763	**0.7013**	0.7327
MOBS	0.5923	0.8038	0.7450	0.7863	0.6838	0.7222
BiMOBS	0.5900	0.8138	0.7637	**0.7887**	0.6913	0.7295
SFCTBS	0.5967	0.8238	0.7488	0.7738	0.6938	0.7274
MGSR	**0.7466**	**0.9262**	**0.8287**	0.7812	**0.7237**	**0.8012**
FNGBS	0.5900	**0.8413**	**0.7938**	0.7762	0.6988	**0.7400**
TOMOBS	0.5933	0.8388	0.7563	**0.7913**	0.6963	0.7352

TABLE V

AUC OF (P_F, τ) VALUES OF THE BAND SUBSET WITH 18 BANDS OF BAND SELECTION METHODS AND FULL BANDS FOR FIVE ROW TARGETS ON HYDICE BY CEM DETECTION

Method	$\mathbf{p}_1$	$\mathbf{p}_2$	$\mathbf{p}_3$	$\mathbf{p}_4$	$\mathbf{p}_5$	Average
Full bands	**0.0328**	**0.1012**	**0.0519**	**0.0546**	**0.0495**	**0.058**
MOBS	0.0574	0.1291	0.1177	**0.0149**	0.0560	0.0750
BiMOBS	0.1601	0.2858	0.1307	0.1374	0.0599	0.1547
SFCTBS	0.0402	0.1578	0.0847	0.0712	**0.0133**	0.0734
MGSR	0.3438	0.4494	0.3746	0.0995	0.1032	0.2741
FNGBS	0.0538	0.1753	0.2689	0.1135	0.0650	0.1353
TOMOBS	**0.0386**	**0.1277**	**0.0782**	0.0649	0.0520	**0.0722**

TABLE VI

AUC OF (P_D, P_F) VALUES OF THE BAND SUBSET WITH 18 BANDS OF BAND SELECTION METHODS AND FULL BANDS FOR FIVE ROW TARGETS ON HYDICE BY ACE DETECTION

Method	$\mathbf{p}_1$	$\mathbf{p}_2$	$\mathbf{p}_3$	$\mathbf{p}_4$	$\mathbf{p}_5$	Average
Full bands	**0.9994**	**0.9998**	**0.9998**	**0.9998**	**0.9998**	**0.9997**
MOBS	0.9992	0.9994	0.9991	**0.9998**	0.9995	0.9994
BiMOBS	**0.9993**	**0.9998**	**0.9996**	**0.9998**	**0.9998**	**0.9996**
SFCTBS	**0.9993**	**0.9997**	**0.9996**	**0.9995**	0.9996	0.9995
MGSR	0.9992	0.9994	0.9991	**0.9998**	0.9995	0.9994
FNGBS	**0.9994**	**0.9997**	0.9996	**0.9998**	**0.9998**	**0.9996**
TOMOBS	**0.9994**	**0.9997**	**0.9996**	**0.9998**	**0.9997**	**0.9996**

TABLE VII

AUC OF (P_D, τ) VALUES OF THE BAND SUBSET WITH 18 BANDS OF BAND SELECTION METHODS AND FULL BANDS FOR FIVE ROW TARGETS ON HYDICE BY ACE DETECTION

Method	$\mathbf{p}_1$	$\mathbf{p}_2$	$\mathbf{p}_3$	$\mathbf{p}_4$	$\mathbf{p}_5$	Average
Full bands	0.6933	0.7988	0.8388	0.7938	0.7488	0.7747
MOBS	0.8633	0.8988	0.8288	0.9163	0.8438	0.8702
BiMOBS	0.8700	0.8838	0.9388	**0.9663**	0.9238	0.9165
SFCTBS	**0.9133**	**0.9238**	**0.9688**	0.8388	0.9013	0.9092
MGSR	0.8633	0.8988	0.8288	0.9163	0.8438	0.8702
FNGBS	**0.9367**	0.8888	0.9538	0.9438	**0.9463**	**0.9338**
TOMOBS	**0.9367**	**0.9463**	**0.9738**	**0.9638**	**0.9263**	**0.9493**

from Table II are shown in Figs. 7 and 8. The ground truths of $\mathbf{p}_1$–$\mathbf{p}_5$ are used for comparisons.

1) Number of Selected Bands Analysis: To show the effectiveness of the VD method in Table I to determine the number of hyperspectral bands, several other numbers of bands are compared here by the AUC values of (P_D, P_F). We select 5, 9, 10, 15, 18, and 20 bands for comparison. As can be seen from Fig. 9, when the number of HYDICE data bands calculated by VD method is $n_{\mathrm{BS}} = n_{\mathrm{VD}} = 9$, the AUC value of the TOMOBS method for CEM detection is higher on the $\mathbf{p}_3$ targets than the adjacent $n_{\mathrm{BS}} = 5$ and $n_{\mathrm{BS}} = 10$. When $n_{\mathrm{BS}} = n_{2VD} = 18$, the AUC value of the TOMOBS method for CEM detection is higher on the $\mathbf{p}_2$–$\mathbf{p}_4$ targets than the adjacent $n_{\mathrm{BS}} = 15$ and $n_{\mathrm{BS}} = 20$. In addition, when the comparison method selects the number of $n_{\mathrm{BS}} = n_{2VD} = 18$ bands, the detection effect is mostly better than the number of other bands, as shown in Figs. 9 and 10, indicating that the VD method is effective in solving the number of hyperspectral bands. Special attention is paid to the fact that because n_{BS} determined by the VD method is underestimated, the number of bands $n_{\mathrm{BS}} = n_{2VD} = 18$ is better for target detection in the MO band selection problem. Therefore, this article recommends using $n_{\mathrm{BS}} = n_{2VD}$ in Table I as the number to guide the selection of bands on HYDICE.

2) 3-D ROC Analysis: Since it is hard to tell visually which method produces the best results of Figs. 7 and 8, three types of 2-D ROC curves refined from the 3-D ROC are adopted to further evaluate detection performance in this article. As stated above, these three types of 2-D ROC curves include 2-D ROC curve of (P_D, P_F), 2-D ROC curve of (P_D, τ), and 2-D ROC curve of (P_F, τ). Tables III–V list the corresponding 2-D ROC values of our method and the other five with 18 selected bands and full bands on HYDICE by CEM detection. Tables VI–VIII list the corresponding 2-D ROC values of our method and the other five ones with 18 selected bands and full bands on HYDICE by ACE detection. Among them, red represents the best value and green represents the second value.

As can be seen, the AUC of (P_D, P_F) on the five row panels of TOMOBS by CEM detection is between 0.9992 and 0.9997, and the AUC value of TOMOBS is almost higher than that of the other two MO algorithms, which is equivalent to that of SFCTBS. This article also adopts 2-D ROC curves of (P_D, τ) and (P_F, τ), which can be jointly used to assess the impact of BKG suppression on target detection. The TOMOBS detection value of AUC of (P_D, τ) is between 0.5933 and 0.8388, and the TOMOBS AUC of (P_D, τ) is the best especially on $\mathbf{p}_4$, even higher than the full bands' value. The TOMOBS detection value of AUC of (P_F, τ) is between 0.0386 and 0.1277, especially on $\mathbf{p}_1$–$\mathbf{p}_3$, the TOMOBS AUC of (P_F, τ) is better than the comparison algorithms. It showed that TOMOBS has better BKG suppression.

To further conduct a quantitative study, Fig. 9(a)–(e) demonstrate the AUC of (P_D, P_F) for CEM detection using 5–20 bands by six BS methods, MOBS, BiMOBS, SFCTBS, FNGBS, MGSR, and TOMOBS, with each of $\mathbf{p}_1$–$\mathbf{p}_5$ on HYDICE used to choose bands to detect panel pixels. Clearly, when the number of bands is between 5 and 20, the AUC value of TOMOBS remains steady between 0.9987 and 0.9997, unaffected by the number of bands.

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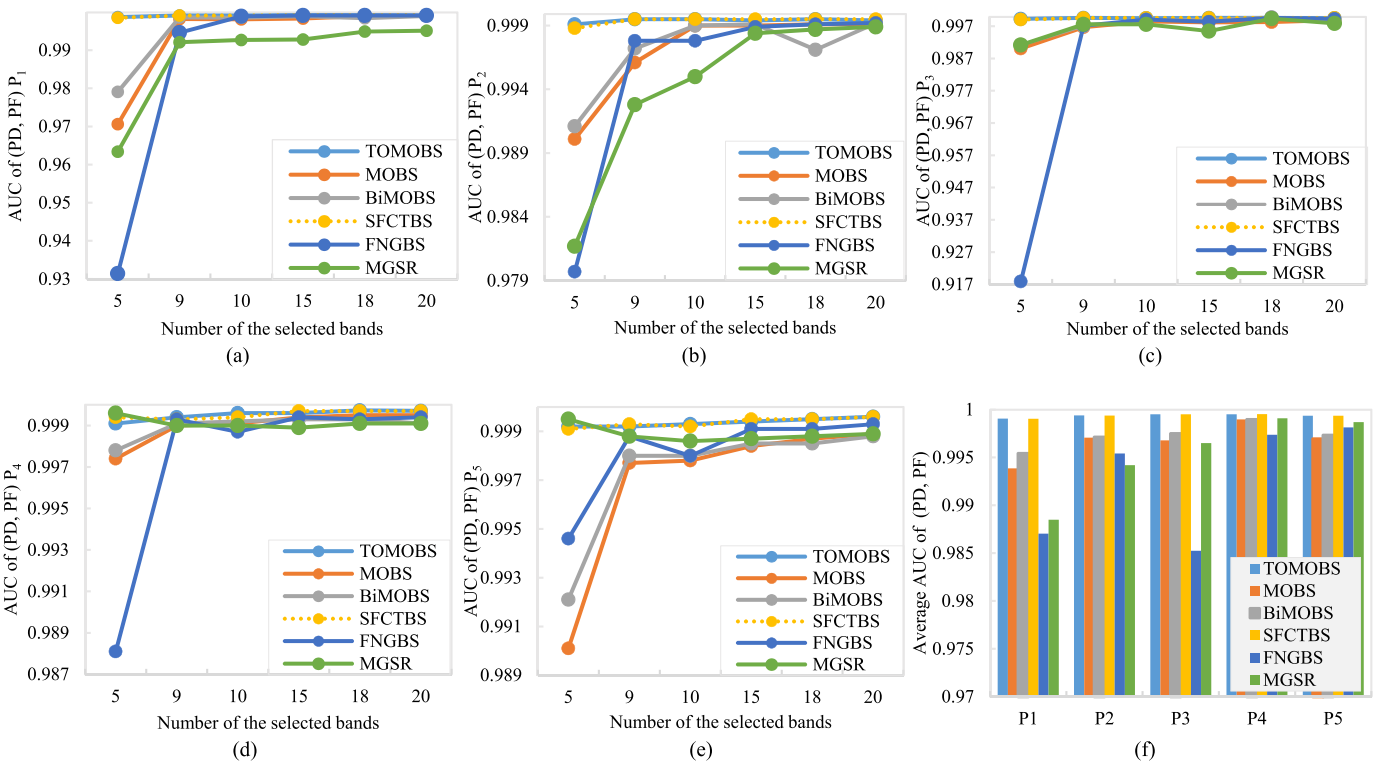


Fig. 9. (a) AUC of (P_D, P_F) for CEM detection results on HYDICE for target $\mathbf{p}_1$. (b) AUC of (P_D, P_F) for CEM detection results on HYDICE for target $\mathbf{p}_2$. (c) AUC of (P_D, P_F) for CEM detection results on HYDICE for target $\mathbf{p}_3$. (d) AUC of (P_D, P_F) for CEM detection results on HYDICE for target $\mathbf{p}_4$. (e) AUC of (P_D, P_F) for CEM detection results on HYDICE for target $\mathbf{p}_5$. (f) Average AUC of (P_D, P_F) for CEM detection results of five row panels using different bands on HYDICE.

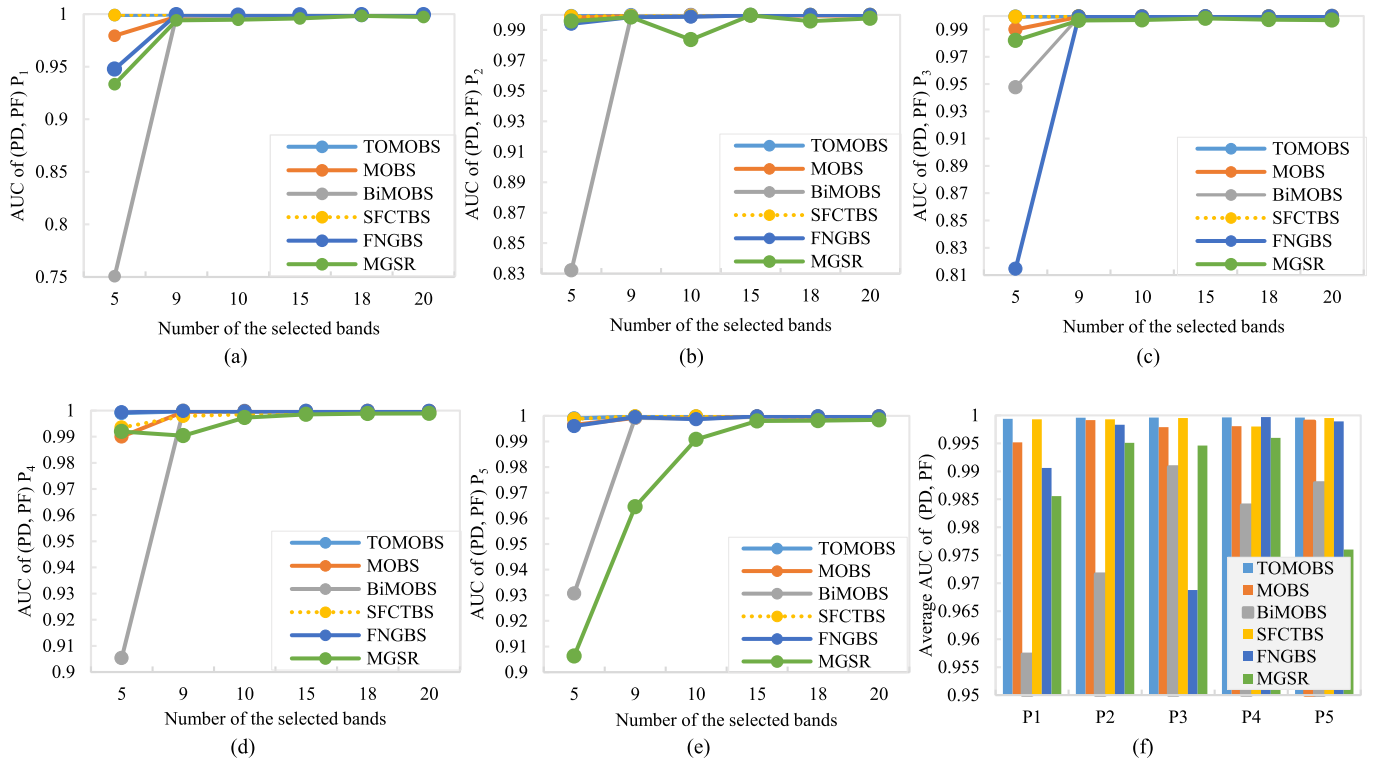


Fig. 10. AUC of (P_D, P_F) for ACE detection results on HYDICE for target (a) $\mathbf{p}_1$. (b) $\mathbf{p}_2$. (c) $\mathbf{p}_3$. (d) $\mathbf{p}_4$. (e) $\mathbf{p}_5$. (f) Average AUC of (P_D, P_F) for ACE detection results of five row panels using different bands on HYDICE.

To better demonstrate the performance of TOMOBS, Fig. 9(f) displays the average AUC of (P_D, P_F) of each method in five row panels with varying band sets. It can be seen that the TOMOBS average AUC of (P_D, P_F) is always nearly 1 on HYDICE, indicating the advantage. When comparing average AUC of (P_D, P_F) results with other algorithms, the TOMOBS

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TABLE VIII

AUC OF (P_F, τ) VALUES OF THE BAND SUBSET WITH 18 BANDS OF BAND SELECTION METHODS AND FULL BANDS FOR FIVE ROW TARGETS ON HYDICE BY ACE DETECTION

Method	$\mathbf{p}_1$	$\mathbf{p}_2$	$\mathbf{p}_3$	$\mathbf{p}_4$	$\mathbf{p}_5$	Average
Full bands	**0.0060**	**0.0060**	**0.0058**	**0.0058**	**0.0057**	**0.0058**
MOBS	0.0230	0.0512	0.0522	0.0200	0.0192	0.0331
BiMOBS	0.0513	0.0493	0.0444	0.0207	0.0217	0.0374
SFCTBS	**0.0157**	0.0380	**0.0141**	0.0198	**0.0158**	**0.0206**
MGSR	0.0230	0.0512	0.0522	0.0200	0.0192	0.0331
FNGBS	0.0224	0.0411	0.0363	0.0256	0.0229	0.0296
TOMOBS	0.0197	**0.0373**	0.0198	**0.0177**	0.0206	0.0230

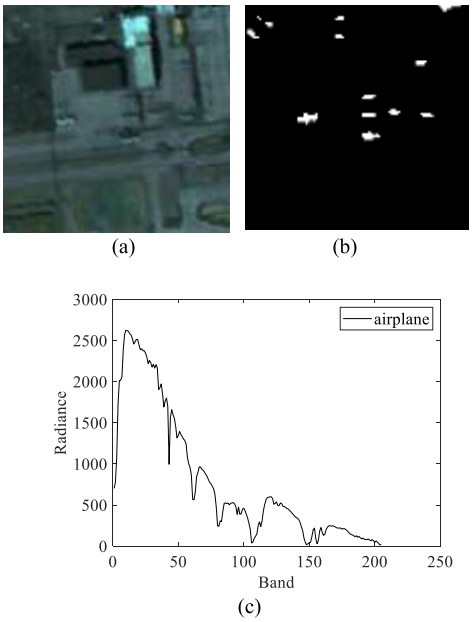


Fig. 11. (a) AVIRIS scene. (b) Ground-truth map. (c) Spectra of the airplane.

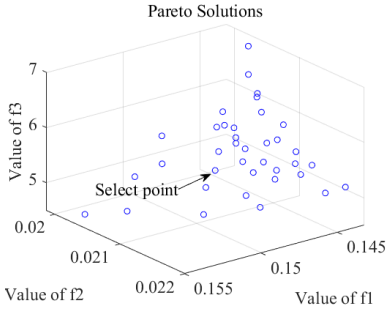


Fig. 12. Three-dimensional PF of the selected band set and TOMOBS method's target selection result for 14 bands on AVIRIS.

algorithm achieves the greatest value. According to the CEM detection findings in Table III, the average AUC of (P_D, P_F) value of SFCTBS is comparable to that of TOMOBS and better than that of other rivals. Because the SFCTBS algorithm also uses desired target signatures by prior knowledge. It shows that using different band subsets for different targets will have better detection effect for target detection. As noted, the AUC of (P_F, τ) reflects the degree of BKG suppression. Table V lists that TOMOBS average AUC of (P_F, τ) is small than SFCTBS, which means that TOMOBS has better BKG suppression. In other words, the detection effect of TOMOBS is better than that of SFCTBS by CEM detection.

TABLE IX

n_{BS} AND $2n_{BS}$ ESTIMATED FOR AVIRIS DATA BY VD

Data Set	n_{BS}	$2n_{BS}$
AVIRIS	14	28

TABLE X

14 BAND SUBSETS SELECTED BY VARIOUS METHODS ON AVIRIS

Method	Band Subset
Full bands	1:1:205
MOBS	105,150,66,84,60,172,138,169,136,35,140,31,184,198
BiMOBS	183,20,105,82,167,125,195,51,27,16,2,56,127,39
SFCTBS	11,159,138,163,154,158,135,3,88,92,84,5,155,86
MGSR	11,10,12,13,14,9,18,19,15,17,20,16,22,21
FNGBS	6,21,33,44,54,81,91,98,121,126,141,165,173,194
TOMOBS	4,22,32,150,57,76,89,178,117,139,144,163,190,181

At the same time, the ACE detection results of 18 bands selected by the proposed algorithm and the comparison algorithms on HYDICE can be seen in Tables VI–VIII. Similar to the CEM detection results, most of the AUC of (P_D, P_F) values of using TOMOBS-selected bands by ACE detection are relatively high, and the TOMOBS AUC of (P_D, P_F) is the best especially on $\mathbf{p}_1$ and $\mathbf{p}_4$. The TOMOBS AUC of (P_D, τ) is the best especially on $\mathbf{p}_1$–$\mathbf{p}_3$. Obviously, the AUC of (P_F, τ) values are relatively low, especially on $\mathbf{p}_2$ and $\mathbf{p}_4$, which means that the background is also well-suppressed while accurately detecting the target. The AUC of (P_D, P_F) values by ACE detection with different numbers of selected bands for different targets are shown in Fig. 10(a)–(e). It can be seen that some AUC of (P_D, P_F) values overlap, indicating that the ACE detection performance of the band selection algorithms is generally good. But it can still be seen that the proposed TOMOBS almost obtains the best values of AUC when selected bands range from 5 to 20 in Fig. 10(a)–(e). Especially in Fig. 10(f), the average AUC of (P_D, P_F) value for each target of the proposed algorithm is significantly higher than that of the comparison algorithms.

The above results show that the TOMOBS method is better than other comparison methods in both CEM and ACE detection, and it also has stable and good detection performance for different targets.

B. AVIRIS Data Results

The second set of data is the hyperspectral remote sensing image data of the airport. The HSI dataset is collected by the AVIRIS in the Los Angeles airport area. AVIRIS contains 205 spectral bands with size of 100 × 100. There are a total of 144 targets in the image of several aircrafts. The image scene and the target reference position are shown in Fig. 11(a) and (b). The spectrum of airplane is shown in Fig. 11(c).

As described previously, to effectively detect the targets, this article adopts the VD method to determine the number of bands. For the AVIRIS dataset, Table IX lists the n_{BS} and $2n_{BS}$ values of VD with a false alarm rate of 10^{-4}.

Table X lists the set of full bands and the 14 bands selected by the six band selection methods, MOBS, BiMOBS, SFCTBS, MGSR, FNGBS, and TOMOBS.

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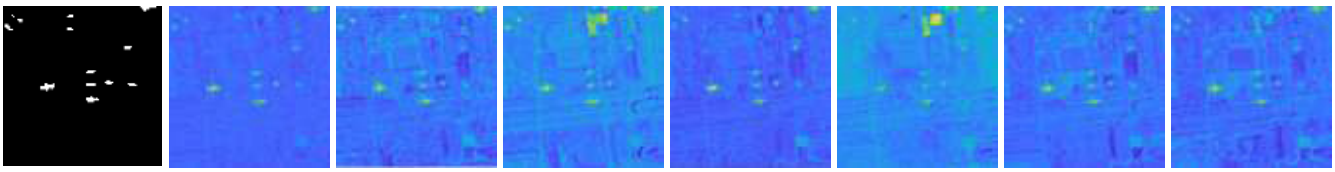


Fig. 13. CEM detection results of 14 band subsets selected by six band selection methods.

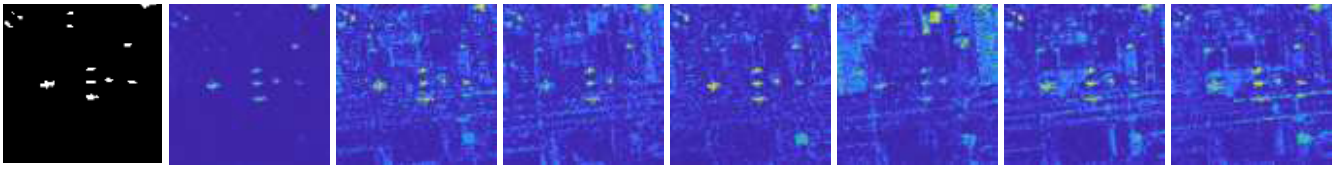


Fig. 14. ACE detection results of 14 band subsets selected by six band selection methods.

Fig. 12 depicts the PF of the selected band set with 14 bands on AVIRIS, which is derived by taking the three objectives into account at the same time. Each of these locations is a subset of the bands and reflects the solution of TOMOBS on AVIRIS.

As shown in Fig. 12, the best points selected by the airplane target are closer to the horizontal axis, which means that on the AVIRIS data, the target reduction of redundancy is more important for detection than high information entropy and low noise effect. The relationship between the three optimization objectives is detailed in Section IV-C.

Fig. 13 shows the detection results of CEM using the band set in Table X. Fig. 14 shows the detection results of ACE. From the detection results in Figs. 13 and 14, it can be concluded that when compared with the bands selected by MOBS, BiMOBS, and MGSR, the bands selected by TOMOBS can better realize the separation of the target from the background. The detailed analysis is as following.

1) Number of Selected Bands Analysis: It can be found from the above experimental results that the number of bands calculated by the VD method in Table X is suitable for band selection on AVRIS data. To further prove the advantages of this method, the number of selected bands will be set from {5,10,14,15,20,25,28,30} below. According to Fig. 15, it can be seen that when the number of AVIRIS data bands calculated by the VD method is $\eta_{\mathrm{BS}} = \eta_{\mathrm{VD}} = 14$, and the AUC of $(P_D,\ P_F)$ value of the TOMOBS method is higher than the adjacent side $\eta_{\mathrm{BS}} = 10$ and $\eta_{\mathrm{BS}} = 15$. Similarly, when $\eta_{\mathrm{BS}} = \eta_{2VD} = 28$, the AUC value of the TOMOBS method is higher than the adjacent $n_{\mathrm{BS}} = 25$ and $n_{\mathrm{BS}} = 30$. In addition, when the comparison method selects the number of bands of $n_{\mathrm{BS}} = n_{\mathrm{VD}}$, the detection effect is also better than the adjacent bands, indicating that the VD method is effective in solving the number of hyperspectral bands.

2) 3-D ROC Analysis: This experiment also compares the AUC of $(P_D,\ P_F)$, AUC of $(P_D,\ \tau)$, and AUC of $(P_F,\ \tau)$ as the HYDICE one. In Table XI, red represents the best value, green represents the second value, and the results verify the above analysis too. As can be seen, the CEM detection value of AUC of $(P_D,\ P_F)$ on the TOMOBS method is greater than that of other methods, even higher than the full bands' value. The AUC of $(P_D,\ \tau)$ and $(P_F,\ \tau)$ on the TOMOBS method is not the best, but TOMOBS ranks the fourth in AUC of $(P_D,\ \tau)$ and the fourth in AUC of $(P_F,\ \tau)$. As noted, the detection value of AUC of $(P_F,\ \tau)$ is better than the other MO methods. At the same time, it can be seen that the TOMOBS method ranks the third in AUC of $(P_D,\ P_F)$ and AUC of $(P_D,\ \tau)$ value detected by ACE, which is superior to the two multiobjective band selection methods. This is sufficient to demonstrate that the detection performance difference between TOMOBS and the comparative technique is considerable. The following findings demonstrate that TOMOBS has considerably greater detection accuracy than other comparative methods, as well as consistent and good detection performance for diverse targets.

TABLE XI

AUC VALUES FOR CEM DETECTION RESULTS OF BAND SUBSET WITH 14 BANDS OF BAND SELECTION METHODS AND FULL BANDS ON AVIRIS

Method	AUC of $(P_{\mathrm{D}}, P_{\mathrm{F}})$	AUC of (P_{D}, τ)	AUC of (P_{F}, τ)
Full bands	**0.9710**	0.4633	**0.2001**
MOBS	0.9383	**0.5165**	0.2373
BiMOBS	0.9395	**0.5374**	0.2967
SFCTBS	0.9428	0.4755	**0.1974**
MGSR	0.8512	0.4614	0.3171
FNGBS	0.9611	0.5047	0.2321
TOMOBS	**0.9721**	0.5005	0.2182

TABLE XII

AUC VALUES FOR ACE DETECTION RESULTS OF BAND SUBSET WITH 14 BANDS OF BAND SELECTION METHODS AND FULL BANDS ON AVIRIS

Method	AUC of $(P_{\mathrm{D}}, P_{\mathrm{F}})$	AUC of (P_{D}, τ)	AUC of (P_{F}, τ)
Full bands	**0.9654**	0.3145	**0.0086**
MOBS	0.9194	0.5241	0.0773
BiMOBS	0.9065	0.4470	0.0752
SFCTBS	**0.9422**	**0.5349**	**0.0568**
MGSR	0.8396	0.3694	0.0987
FNGBS	0.9199	**0.5332**	0.0796
TOMOBS	0.9232	0.5247	0.0798

Fig. 15(a) depicts the AUC of $(P_D,\ P_F)$ for CEM identification using 5–30 bands by six approaches on AVIRIS: MOBS, BiMOBS, SFCTBS, FNGBS, MGSR, and TOMOBS. Clearly, when the number of bands is between 5 and 30, the AUC value of TOMOBS is constant between 0.91 and 0.988, which is unaffected greatly by the number of bands, and the AUC value of TOMOBS is the best. To better

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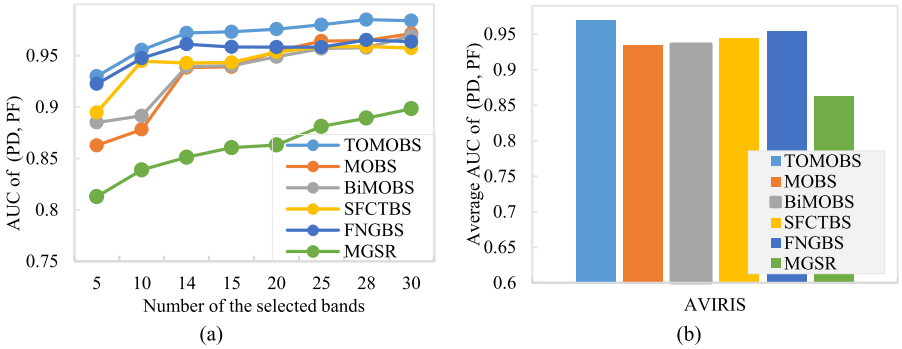


Fig. 15. (a) AUC of (P_D, P_F) for CEM detection results on AVIRIS. (b) Average AUC of (P_D, P_F) for CEM detection results of different bands on AVIRIS.

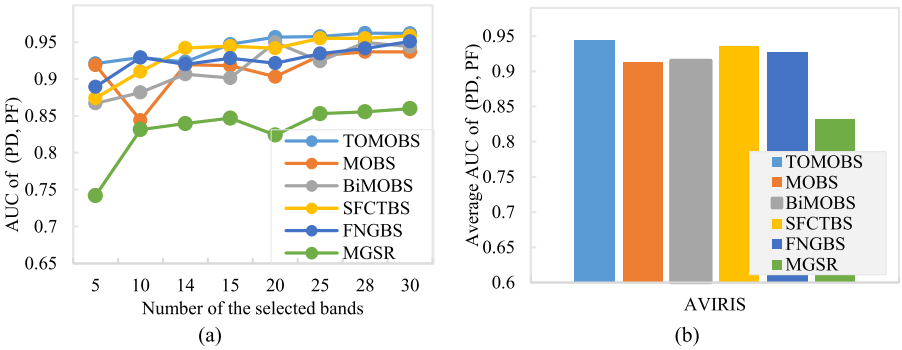


Fig. 16. (a) AUC of (P_D, P_F) for ACE detection results on AVIRIS. (b) Average AUC of (P_D, P_F) for ACE detection results of different bands on AVIRIS.

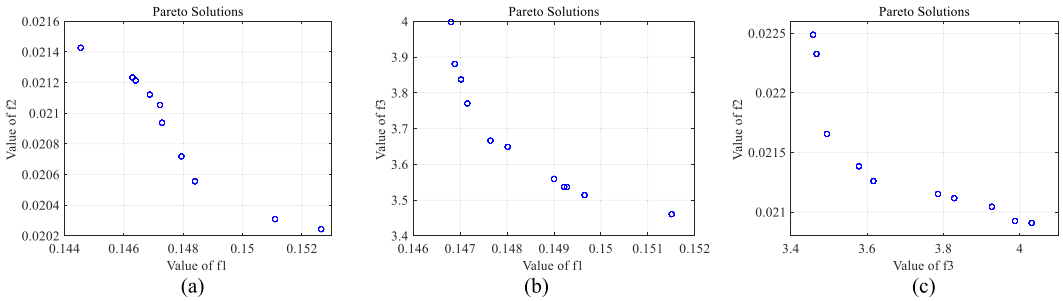


Fig. 17. (a) PF of 18 band subsets selected by $f1$ and $f2$ objective functions on HYDICE. (b) PF of 18 band subsets selected by $f1$ and $f3$ objective functions on HYDICE. (c) PF of 18 band subsets selected by $f2$ and $f3$ objective functions on HYDICE.

demonstrate the performance of TOMOBS, Fig. 15(b) displays the average value of AUC of (P_D, P_F) for each method distinct band numbers. TOMOBS has the highest value based on the AUC average findings when compared with other competitors.

As previously stated, the average AUC value of SFCTBS is similar to that of TOMOBS and better than that of other rivals based on CEM detection results. However, the AUC detection value of the SFCTBS method under different band numbers is not better than TOMOBS. As previously stated, the TOMOBS technique outperforms other comparison methods in terms of detection accuracy and has a steady and good detection performance.

At the same time, the ACE detection results of 14 bands selected by the proposed algorithm and the comparison algorithms on AVIRIS can be seen in Figs. 14 and 16. From Fig. 16(f), the average AUC of (P_D, P_F) value for each target of the proposed algorithm is also higher than that of the comparison algorithms.

The above results show that the TOMOBS method is better than other comparison methods in detection accuracy, and it also has stable and impressive performance for the target.

C. Ablation Experiments

To better reflect the relationship between the three optimization objectives of TOMOBS, we perform ablation experiments on the HYDICE and AVIRIS datasets. In the experiment, the band subset is selected by combining two optimization objectives and TOMOBS. The PF of 18 band subsets is selected by different objective functions on HYDICE, as shown in Fig. 17. The CEM detection of five row panels using 18 bands selected by TOMOBS and different objective functions on HYDICE in Table XIII are shown in Fig. 18. Similarly, the PF and the CEM detection results of AVIRIS dataset are shown in Figs. 19 and 20. In the table, red represents the best value.

The following conclusions can be drawn from Tables XIV–XVI.

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TABLE XIII
18 BAND SUBSETS SELECTED BY DIFFERENT OBJECTIVE FUNCTIONS ON HYDICE

	Method	Band Subset
$\mathbf{p}_1$	f1 and f2	103,75,80,57,59,127,42,53,126,143,87,134,50,40,6,20,100,45
$\mathbf{p}_1$	f1 and f3	144,132,66,45,22,152,2,68,74,111,130,58,95,19,114,36,96,53
$\mathbf{p}_1$	f2 and f3	1,13,22,37,44,54,62,68,79,90,96,107,112,123,135,136,146,157
$\mathbf{p}_1$	TOMOBS(f1f2f3)	3,17,28,128,64,122,57,162,106,97,90,22,112,33,133,140,150,157
$\mathbf{p}_2$	f1 and f2	103,75,80,57,59,127,42,53,126,143,87,134,50,40,6,20,100,45
$\mathbf{p}_2$	f1 and f3	144,132,66,45,22,152,2,68,74,111,130,58,95,19,114,36,96,53
$\mathbf{p}_2$	f2 and f3	1,13,22,37,44,54,62,68,79,90,96,107,112,123,135,136,146,157
$\mathbf{p}_2$	TOMOBS(f1f2f3)	143,6,19,26,30,121,41,57,68,129,153,76,82,96,100,111,132,146
$\mathbf{p}_3$	f1 and f2	103,75,80,57,59,127,42,53,126,143,87,134,50,40,6,20,100,45
$\mathbf{p}_3$	f1 and f3	144,132,66,45,22,152,2,68,74,111,130,58,95,19,114,36,96,53
$\mathbf{p}_3$	f2 and f3	1,13,22,37,44,54,62,68,79,90,96,107,112,123,135,136,146,157
$\mathbf{p}_3$	TOMOBS(f1f2f3)	136,96,27,33,140,52,90,70,82,68,146,105,54,127,37,159,15,46
$\mathbf{p}_4$	f1 and f2	37,14,69,18,19,26,28,63,50,61,138,152,84,80,103,35,15,122
$\mathbf{p}_4$	f1 and f3	44,132,66,45,22,152,2,68,74,111,130,58,95,19,114,36,96,53
$\mathbf{p}_4$	f2 and f3	1,13,22,37,44,54,62,68,79,90,96,107,112,123,135,136,146,157
$\mathbf{p}_4$	TOMOBS(f1f2f3)	99,50,26,32,141,54,60,71,110,84,105,100,111,114,126,139,147,156
$\mathbf{p}_5$	f1 and f2	52,87,107,54,69,3,102,82,64,114,75,125,77,104,42,120,132,57
$\mathbf{p}_5$	f1 and f3	144,132,66,45,22,152,2,68,74,111,130,58,95,19,114,36,96,53
$\mathbf{p}_5$	f2 and f3	1,13,22,37,44,54,62,68,79,90,96,107,112,123,135,136,146,157
$\mathbf{p}_5$	TOMOBS(f1f2f3)	132,113,123,25,101,136,99,124,14,38,115,165,28,151,52,90,109,119

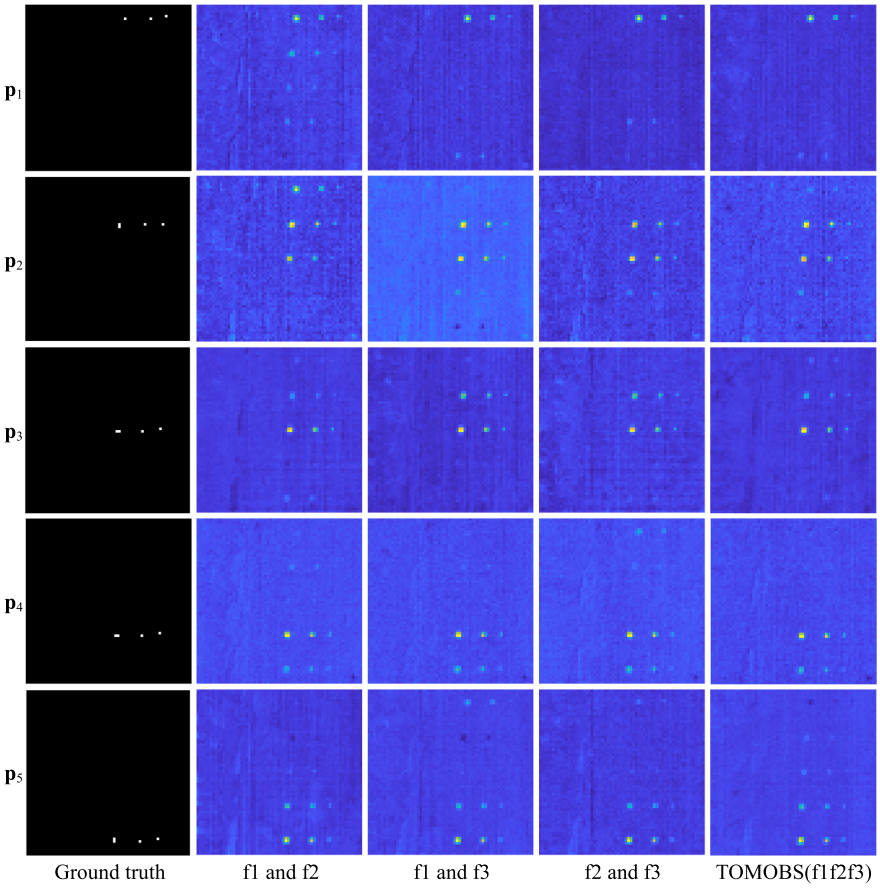


Fig. 18. CEM detection of five row panels using 18 bands selected by TOMOBS and different objective functions on HYDICE in Table XIII.

First, on the HYDICE data, the AUC of (P_D, P_F) of the three objective functions is higher than the values of any two objective functions, indicating that the three objective functions have better detection effect. The AUC of (P_F, τ) values of the three objective functions are almost lower than the values of any two objective functions, which means that the three objective functions have better BKG suppression.

Second, on the HYDICE data, $f1$ and $f2$ play a more important role of the $\mathbf{p}_1$, $\mathbf{p}_3$, and $\mathbf{p}_5$ targets in target detection. $f3$ effect plays a more important role in the detection of the $\mathbf{p}_2$ target. For the $\mathbf{p}_4$ target, $f1$, $f2$, and $f3$ effects play an equally important role. It can be concluded that for different targets, different objective functions have different effects. Therefore, for the HYDICE dataset, it is better to adopt three objective functions.

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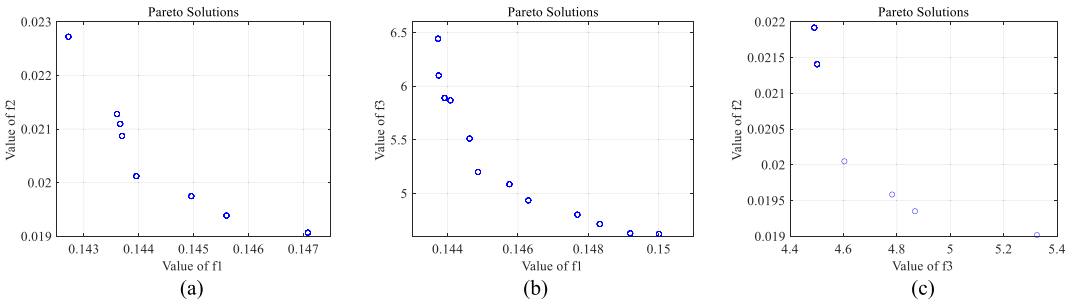


Fig. 19. (a) PF of 14 band subsets selected by $f1$ and $f2$ objective functions on AVIRIS. (b) PF of 14 band subsets selected by $f1$ and $f3$ objective functions on AVIRIS. (c) PF of 14 band subsets selected by $f2$ and $f3$ objective functions on AVIRIS.

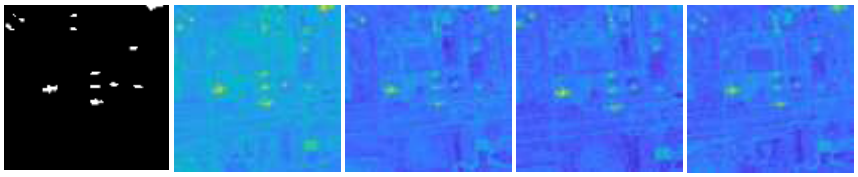


Fig. 20. CEM detection results of 14 band subsets selected by different objective functions on AVIRIS in Table XVII.

TABLE XIV

AUC OF (P_D, P_F) VALUES OF THE BAND SUBSET WITH 18 BANDS OF DIFFERENT OBJECTIVE FUNCTIONS AND TOMOBS FOR FIVE ROW TARGETS ON HYDICE BY CEM DETECTION

Method	$\mathbf{p}_1$	$\mathbf{p}_2$	$\mathbf{p}_3$	$\mathbf{p}_4$	$\mathbf{p}_5$
f1 and f2	0.9987	0.9988	**0.9996**	0.9994	0.9989
f1 and f3	0.9991	0.9990	0.9993	0.9993	0.9988
f2 and f3	**0.9992**	0.9984	0.9991	0.9993	0.9987
TOMOBS(f1f2f3)	**0.9992**	**0.9995**	**0.9996**	**0.9997**	**0.9995**

TABLE XV

AUC OF (P_D, τ) VALUES OF THE BAND SUBSET WITH 18 BANDS OF DIFFERENT OBJECTIVE FUNCTIONS AND TOMOBS FOR FIVE ROW TARGETS ON HYDICE BY CEM DETECTION

Method	$\mathbf{p}_1$	$\mathbf{p}_2$	$\mathbf{p}_3$	$\mathbf{p}_4$	$\mathbf{p}_5$
f1 and f2	**0.6233**	0.7938	0.7388	0.7837	**0.7138**
f1 and f3	0.6067	0.8250	**0.7663**	0.7813	0.7038
f2 and f3	0.6100	0.6725	0.7263	0.7962	0.6913
TOMOBS(f1f2f3)	0.5933	**0.8388**	0.7563	**0.7913**	0.6963

TABLE XVI

AUC OF (P_F, τ) VALUES OF THE BAND SUBSET WITH 18 BANDS OF DIFFERENT OBJECTIVE FUNCTIONS AND TOMOBS FOR FIVE ROW TARGETS ON HYDICE BY CEM DETECTION

Method	$\mathbf{p}_1$	$\mathbf{p}_2$	$\mathbf{p}_3$	$\mathbf{p}_4$	$\mathbf{p}_5$
f1 and f2	0.0969	0.1110	0.0849	0.1333	0.0874
f1 and f3	0.0756	0.1962	**0.0699**	0.1279	0.1024
f2 and f3	0.0567	**0.0990**	0.0897	0.1399	0.0858
TOMOBS(f1f2f3)	**0.0386**	0.1277	0.0782	**0.0649**	**0.0520**

On the AVIRIS data, it can be seen in Table XVIII that the AUC values of (P_D, P_F) of the three objective functions are higher than the values of any two objective functions, indicating that the three objective functions have better detection effect. The AUC values of (P_F, τ) of the three objective functions are almost lower than the values of any two objective functions, which means that the three objective functions have better BKG suppression.

TABLE XVII

14 BAND SUBSETS SELECTED BY DIFFERENT OBJECTIVE FUNCTIONS AND TOMOBS ON AVIRIS

Method	Band Subset
f1 and f2	15,17,33,46,64,85,91,113,117,136,144,167,179,186
f1 and f3	2,24,42,52,61,81,97,102,117,129,142,169,180,186
f2 and f3	143,12,102,31,50,127,76,9,183,42,17,188,144,168
TOMOBS(f1f2f3)	4,22,32,150,57,76,89,178,117,139,144,163,190,181

TABLE XVIII

AUC VALUES FOR CEM DETECTION RESULTS OF BAND SUBSET WITH 14 BANDS OF DIFFERENT OBJECTIVE FUNCTIONS AND TOMOBS ON AVIRIS

Method	AUC of (P_D, P_F)	AUC of (P_D, τ)	AUC of (P_F, τ)
f1 and f2	0.9546	**0.6266**	0.3758
f1 and f3	0.9234	0.4833	0.2377
f2 and f3	0.9593	0.5033	0.2192
TOMOBS(f1f2f3)	**0.9721**	0.5005	**0.2182**

The above two data results show that using the three objective functions of information entropy, SNR estimation, and JSD has better detection effect than using two of them alone.

D. Band Clarity Analysis

To reflect the clarity of the bands selected by each method, this article uses the AVIRIS data to analyze SNR estimation of selected band subsets with ten bands, as shown in Fig. 21. As mentioned above, the higher the noise, the lower the SNR estimation value. By comparing SNR estimation of the ten band subsets of each method of the AVIRIS data, we find that the band selected by the TOMOBS method is concentrated in the upper middle position, and the selected bands have high SNR estimation, indicating that that the selected bands have low noise. The band selected by other band selection algorithms is concentrated in the lower middle position, and SNR estimation of the selected bands is generally low, indicating that he selected bands have high noise.

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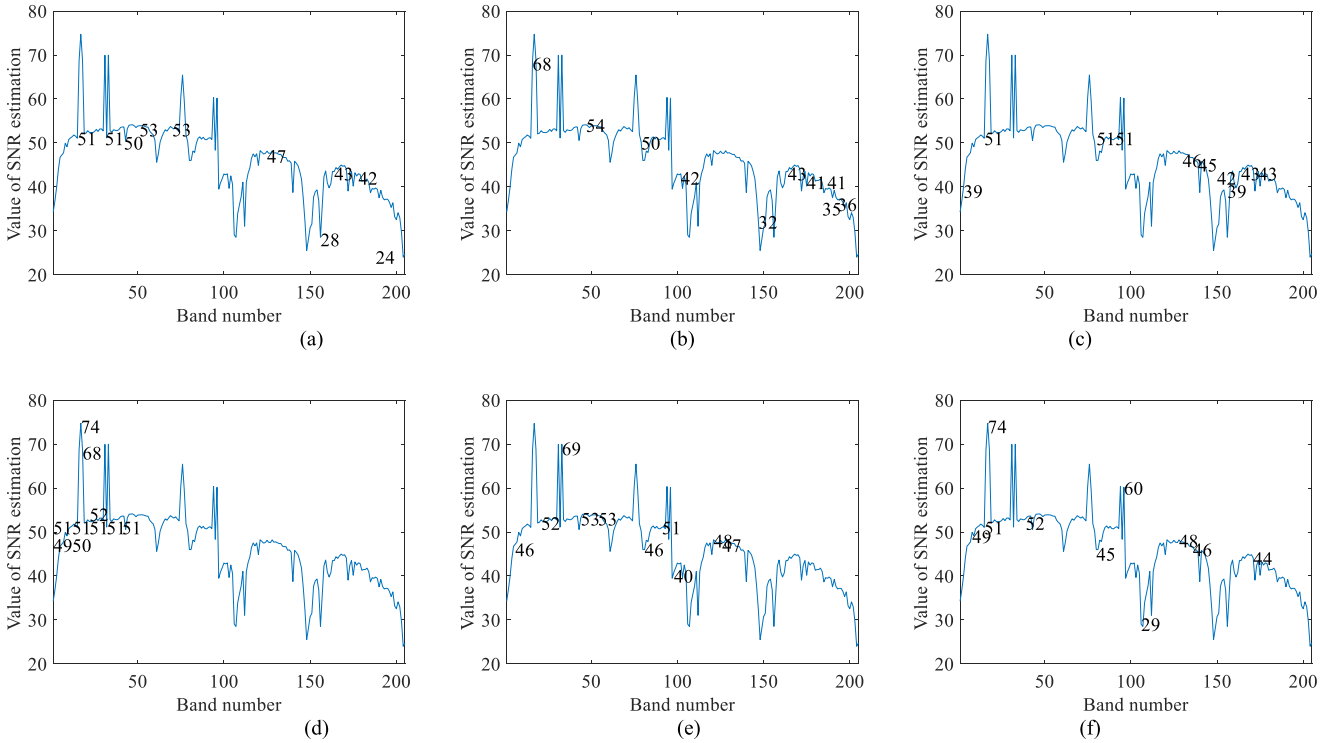


Fig. 21. SNR estimation of selected band subsets with ten bands on AVIRIS. (a) MOBS. (b) BiMOBS. (c) SFCTBS. (d) MGSR. (e) FNGBS. (f) TOMOBS.

TABLE XIX

TEN BAND SUBSETS SELECTED BY VARIOUS METHODS ON AVIRIS AND AVERAGE SNR ESTIMATION

Method	Average SNR estimation
MOBS	44.57
BiMOBS	44.56
SFCTBS	45.45
MGSR	**54.8**
FNGBS	50.5
TOMOBS	**50.59**

To show the clarity of the selected bands, this article uses the average SNR estimation of the band subsets for comparison, as shown in Table XIX. Among them, red represents the highest value and green represents the second value. Among the six methods, the value of the TOMOBS method ranks second, which is obviously better than the other two multiobjective band selection algorithms, indicating that the selected band has low noise. Although the average SNR estimate of the MGSR algorithm is the highest, the selected bands are mainly concentrated between 1 and 20 bands, which means that there is a large redundancy in the selected bands. Combined with the AVIRIS detection results in Fig. 15, the TOMOBS method has the highest average AUC value among the six methods. It can be concluded that selecting bands with low noise is expected to improve detection.

E. Analysis of Computational Time Complexity

To record the run time of band selection, the experiment is performed on a computer equipped with an Intel Core i5-8400 2.8-GHz CPU and 8-GB RAM. The methods

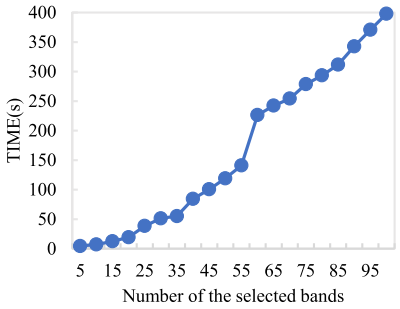


Fig. 22. Execution time of TOMOBS when the number of the selected bands ranges from 5 to 100 on HYDICE.

TABLE XX

CALCULATION TIME (IN SECONDS) FOR SELECTING TEN BANDS

Method	HYDICE	AVIRIS
MOBS	6.9132	7.6499
BiMOBS	7.1593	8.2345
SFCTBS	1.7243	2.2055
MGSR	**1.3061**	**2.0606**
FNGBS	**1.6889**	**2.5451**
TOMOBS	5.1336	6.2542

MOBS, BiMOBS, TOMOBS, SFCTBS, MGSR, and FNGBS are implemented in MATLAB R2021a.

The execution times of these five methods are listed in Table XX. Among them, red represents the best value and green represents the second value. The number of iterations of the MO method is uniformly set to 50 times, and the population size is the same.

In terms of time, the MGSR method outperforms all the other methods, despite the fact that its detection performance is unstable and not satisfactory on both the datasets. Since

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TABLE XXI
COMPARISON OF DIFFERENT NUMBERS OF BAND SUBSETS SELECTED BY TOMOBS ON HYDICE

	5 Bands	10 Bands	15 Bands	20 Bands
$\mathbf{p}_1$	5,118,140,36,162	3,128,61,102,57,162,106,133,111,150	17,133,128,110,122,57,162,157,97,140,90,27,64,33,102	43,133,107,78,97,46,64,30,93,162,6,128,22,50,155,102,57,112,140,15
$\mathbf{p}_2$	112,68,153,144,136	6,133,149,129,82,22,111,104,122,153	4,19,26,31,104,129,76,81,97,107,153,121,132,144,111	109,26,30,143,40,57,68,129,76,120,6,82,139,97,111,132,146,15,107,12
$\mathbf{p}_3$	51,143,105,137,45	137,43,141,105,97,70,117,120,3,51	130,97,127,51,143,85,71,3,105,138,20,120,159,115,41	137,6,17,33,140,52,97,70,85,78,50,105,51,120,37,159,15,46,10,130
$\mathbf{p}_4$	79,105,55,114,149	156,110,89,105,90,50,119,126,139,79	112,24,126,31,156,141,68,99,105,71,110,108,139,87,143	141,26,54,60,70,110,84,105,109,111,50,97,9,39,114,126,139,147,32,156
$\mathbf{p}_5$	113,25,16,52,92	129,105,11,25,47,114,68,151,59,81	73,14,105,51,146,132,28,99,38,119,115,165,152,91,109	119,76,40,129,105,90,136,59,124,47,31,15,165,28,113,154,52,109,2,144

TABLE XXII
COMPARISON OF DIFFERENT NUMBERS OF BAND SUBSETS SELECTED BY TOMOBS ON AVIRIS

5 Bands	10 Bands	15 Bands	20 Bands
33,176,110,69,141	116,184,32,101,176,139,191,85,144,110	178,156,33,181,51,23,76,191,89,5,116,139,163,104,142	183,104,51,5,140,176,165,132,101,79,116,150,21,191,89,145,139,70,45,33

MGSR and FNGBS do not require loop operation, they take less time. Compared with methods that require iteration, the TOMOBS method has the shortest execution time among the MO band selection methods. Given that the TOMOBS method can provide better detection performance in the target detection, the running time is acceptable.

Fig. 22 depicts the TOMOBS execution timeline when the number of bands is from 5 to 100. From a practical point of view, the implementation of TOMOBS is effective when the number of selected frequency bands is relatively small, particularly 5–55 bands. Furthermore, TOMOBS can be speeded up by parallel processing in the step of solving MO, which means it could be a practical candidate algorithm.

F. Analyze Band Overlap

For the same target, there will be overlap between band subsets in different sizes. Comparison of the band subsets in different sizes selected by TOMOBS on HYDICE and AVIRIS is shown in Tables XXI and XXII, where the same band is underlined.

From Tables XXI and XXII, different sizes of band subsets selected by TOMOBS are not exactly the same, but they are very similar. This is because the TOMOBS algorithm uses the swarm intelligence optimization method to solve the MO process, gradually approaches the optimal solution. The randomization of the initial solution, the difference in the number of iterations, and the mutation operation may affect the final solution, so their results are not necessarily the same. Theoretically, when the number of iterations is large enough, the final result will be the same. However, as the number of iterations increases, the computing time increases geometrically, as shown in Fig. 22.

To clearly show the overlap degree between the band subsets for the same target, we count the number of overlap bands in different sizes of band subsets selected by TOMOBS on HYDICE and AVIRIS. From Table XXIII, the overlapping number of frequency bands 5, 10, and 15 on HYDICE accounts for half of the total and 20 bands with one-third of the total. From Table XXIV, the overlapping number of bands 5 and 10 on AVIRIS accounts for half of the total, and 15 and 20 bands nearly half of the total. In addition, when the subset gets large, the number of overlap bands may not reach half, but there are many similar bands.

TABLE XXIII
OVERLAP NUMBER OF DIFFERENT NUMBERS OF BAND SUBSETS SELECTED BY TOMOBS ON HYDICE

	5 Bands	10 Bands	15 Bands	20 Bands
$\mathbf{p}_1$	2	5	8	8
$\mathbf{p}_2$	2	5	8	8
$\mathbf{p}_3$	4	6	7	8
$\mathbf{p}_4$	3	6	6	8
$\mathbf{p}_5$	3	5	5	10

TABLE XXIV
OVERLAP NUMBER OF DIFFERENT NUMBERS OF BAND SUBSETS SELECTED BY TOMOBS ON AVIRIS

5 Bands	10 Bands	15 Bands	20 Bands
3	6	7	9

According to Figs. 9 and 15, the AUC of (P_D, P_F) values of different bands on HYDICE dataset is between 0.992 and 0.9997, and the AUC of (P_D, P_F) values of different bands on the AVIRIS dataset is between 0.91 and 0.988. It means that the bands selected by TOMOBS contribute to excellent performance and stability. Although the band subsets in different sizes are slightly different, nearly half of the bands overlap, and the performance of detecting the same target is stable and effective.

Briefly, the experimental findings reveal that the proposed MO model with three objective functions is crucial in finding effective band subsets. Meanwhile, the NAM based on the swarm intelligence optimization method is useful to avoid local optimization. In the obtained PF, the target-oriented evaluation mechanism is important for guiding the selection of the final solution. Therefore, TOMOBS can obtain equal or better detection performance among the comparison algorithms in target detection.

V. CONCLUSION

In this article, a new MO band selection method for HSIs, TOMOBS, is proposed. It uses a new MO framework with

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three conflicting objective functions, including information entropy, SNR estimation, and JSD, to assess the bands more completely. Because the noise of the bands is considered in the objective functions, the influence of high-noise bands on band selection is released. Furthermore, NAM is designed to avoid the swarm optimization falling into local optimum, and the relationship in the vector space of the population is described to overcome the problem of weak interpretability of MO. Moreover, the target-oriented evaluation mechanism is newly established as the TOMOBS selection standard to select the final result from PF. Based on the target-oriented characteristics, the band subset selected by TOMOBS can obtain better detection accuracy in target detection. The experimental results of target detection show that the three objective functions' method proposed in this article is mainly better than the other band selection methods. Moreover, we study the determination method for number of bands, where VD algorithm is believed to be a convincing one.

Although this technique has a good dimensionality reduction ability for HSIs, when used to the practical engineering area, the swarm intelligence optimization algorithm does need time cost. In this scenario, we will optimize the swarm intelligence algorithm to minimize time complexity. In addition, we will investigate the function that is appropriate for multiple-target detection in the future research.

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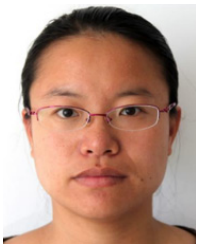
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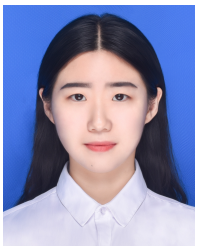
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